



NanoStar Software

Service & Acceptance Macros

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Manual for Service and Acceptance Macros



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Version

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Service & Acceptance Macros

“Service Macros” cover a set of macros to be helpfully used during installation or maintenance of MueTec measurement and inspection systems for masks and wafers.

Only users with software excess level up from “service” are allowed to work with these macros.

Additionally the “Acceptance Macro” exists for certifying the performance of the systems.

The macros are programmed to be exclusively used with the special Leica & MueTec Test Mask or with a MueTec test wafer having an identical layout as the mask has. This layout consists of lines & spaces, dots and holes, arranged in a definite pattern over the substrate.

Mask size can be 5 or 6 inch, wafers of 6 and 8 inch are allowed, depending from the stage driving range.

The macros described here work with NanoStar software up from version s412.

Though it is recommended to use the macros for all system-adjustments and calibrations which they support, it is also possible to use only some macros of the set for selected tasks.

As these service macros widely use the configuration-functionality of the software – other than the former service macros did – there are only some special parameters which have to be learned for exclusive use in context with these macros. Most of these special parameters are related to the mask-layout and the holder.

The tool parameters set by use of the service macros will be valid (as defaults) for the system. They are saved in the system ini files of the NanoStar software.

Basic requirements

Before starting any work with the service macros the safety hardware switch at the microscope should be carefully set to prevent a crash between an objective and the mask.

The pitch calibration of the objectives should be roughly valid. Otherwise the alignments and ROI settings – used in the macros - will not work properly. If necessary, do manually a rough calibration in x and y for each objective and path and wavelength before starting the service macros. (Execute a CD measurement at a quite well known line. The numbers near by the structures on the test substrates stand for the line width of lines and spaces. It is good for rough calibration, to measure simply with LFS or on a manually focused image. Use then CD Window → File → Calibrate x/y axis. Save calibration before rotating to the next objective.)

The Installation path of the service macros is freely selectable. By default it is assumed they are stored in c:\!service\service.app. The “chain”-folder containing some special files has to be a direct subdirectory of the main directory.

The path for the acceptance macro is also freely selectable. By default it is c:\!service\tools.app. Important: Make sure that the complete content of the application directory will be transported to the alternative application directory, if such one should be used.

Number of service macros

Currently 12 macros belong to the set of service macros.

If a tool is adjusted for the first time it is recommended to use the macros in the order of their numbers.

Macro 000 should be used as the last one. It offers several functionalities to get an overview on all valid parameters.

The existing macros do not yet support the adjustment of a robot with a wafer machine. Probably some additional macros for use with wafer tools will be offered in future.

List of the service macros

00_configuration.mac
01_holder_and_layout.mac
02_lenspos_manually.mac
03_illumination.mac
04_planarity.mac
05_revolver_offset.mac
06_LFSspot.mac
07_lenspos_automatically
08_pitch_calibration.mac
09_stage_correction.mac
10_pixel_shading.mac
11_parameter_overview.mac

Universal remarks

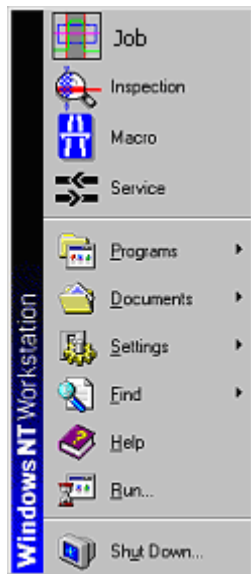
All macros will display the Docu Window on the screen after their start, giving an overview on the function keys available and some very important hints. In its first line the system type and the version number of the macro is displayed. The second line shows the macro name.

Some keys have the same functionality in each macro:

F1:	Refreshes the start screen.
F2:	Executes a stage alignment on the test substrate, which is nearly identically for all macros.
F8:	Serves for substrate loading or unloading, harking back to the Holder Window or the Handler Window. If a valid holder is not yet defined one should handle the mask manually – and carefully pay attention not to damage either the mask or the tool. The same in sense is to be said if the robot is not yet adjusted to the tool
F10:	This is an optional alternative to “Start à Service” to select and start the next service macro. Before using the System’s Start button one has to define in the Applic Window the directory where the service macros are stored. If a macro was loaded by the “File Open” function of the Macro Window the command “Exec” has to be used to start the macro (EXEC button). Otherwise an error message can occur as the application directory is not properly defined. Consequently all “file open” or “file save” commands will not be executed properly.
S1 (=Shift + F1)	This key opens the online help for the service macros. The actually active section will be that one related to the macro from where the help was opened.

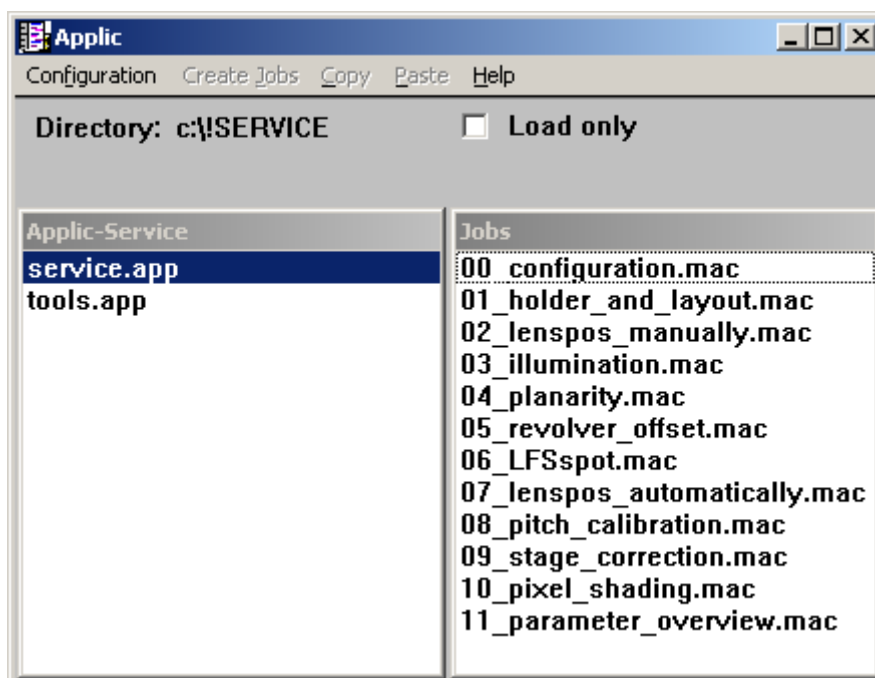
As the tool parameters which are defined using the service macros will be commonly valid for the complete software it is very important to select and set them very carefully.

To start the service macros click the Start Button  on the screen and then click at Service:



The Applic Window similar as shown below will appear on the screen (before the path where the service macros are saved must be defined via Applic Window à Configuration).

Select the wanted macro by a click. It will immediately start.



Description of the macros and their capability

00 Configuration Macro



This is a macro to do the basic system settings. After all functions were executed the system is ready for further, more sophisticated adjustments and calibrations.

F2:

In a dialog box the actually used system should be selected. This name will be displayed in all Service macros in the Docu Window, first line's, upper left corner.

F3:

The dialog box from Config Window à Microscope à Objectives comes up to edit the objective parameters.

Obj	Type	Magnification	N.A.	LWD	Lfs	Camera
1	PL FL	2.5	0.07	☒	☒	1 2 3 4 5 ☒ ☐ ☐ ☐ ☐
2	PL FL B/D	5.0	0.15	☒	☒	1 2 3 4 5 ☐ ☒ ☐ ☐ ☐
3	PL FL B/D	10.0	0.30	☒	☒	1 2 3 4 5 ☒ ☐ ☐ ☐ ☐
4	HC FL B/D	10.0	0.40	☒	☒	1 2 3 4 5 ☐ ☒ ☐ ☐ ☐
5	PL FL L	63.0	0.70	☐	☒	1 2 3 4 5 ☒ ☐ ☐ ☐ ☐
6	PL APO B/D	100.0	0.90	☐	☒	1 2 3 4 5 ☐ ☒ ☐ ☐ ☐

Read actual parameters

Save Cancel

Check carefully the check boxes enabling the use of LFS (automatic laser focus system) for the objectives and the working distance. If not correctly set error messages will probably and unexpectedly brake some commands used later. [E.g. the Holder Window does only accept objectives which are checked for LWD (= long working distance) to be used during verify or exchange operations. If LFS is not checked no measurements with LFS will be executed.]

For tools using multiple cameras that are adapted to different illuminations you should define also the camera to be used with each objective.

Continue with "Save"

F4:

The dialog box from Config Window → Microscope → Illumination comes up to define the wavelengths and ray paths of illumination for each objective. What was not checked here can not be used later on. If both ray paths (reflected and transmitted) are principally possible but only one should be used – check only that one. Do so with the illumination types, too. But remember: what is not allowed here can not be used later.

Microscope Illumination Configuration TL: IR [X]

Obj		vis	I-line	DUV	Infrared
1	Refl	☐	☐	☐	☒
2.5L	Trans	☐	☐	☐	☐
2	Refl	☒	☐	☐	☐
5L	Trans	☐	☐	☐	☐
3	Refl	☐	☐	☐	☒
10L	Trans	☐	☐	☐	☐
4	Refl	☒	☐	☐	☐
10L	Trans	☐	☐	☐	☐
5	Refl	☐	☐	☐	☒
63	Trans	☐	☐	☐	☐
6	Refl	☒	☐	☐	☐
100	Trans	☐	☐	☐	☐

Save Cancel

Continue with "Save"

The next dialog box is that one from Config Window à Microscope à Apertures. It offers only those objectives which were allowed for use with reflected light path.

Microscope Aperture Settings TL: IR [X]

			1	2	3	4	5	Auto change
Obj. 1	2.5x IR	Reflect	☒	☐	☐	☐	☐	☒
Obj. 2	5x Vis	Reflect	☒	☐	☐	☐	☐	☒
Obj. 3	10x IR	Reflect	☒	☐	☐	☐	☐	☒
Obj. 4	10x Vis	Reflect	☒	☐	☐	☐	☐	☒
Obj. 5	63x IR	Reflect	☒	☐	☐	☐	☐	☒
Obj. 6	100x Vis	Reflect	☐	☐	☒	☐	☐	☒

Save Cancel

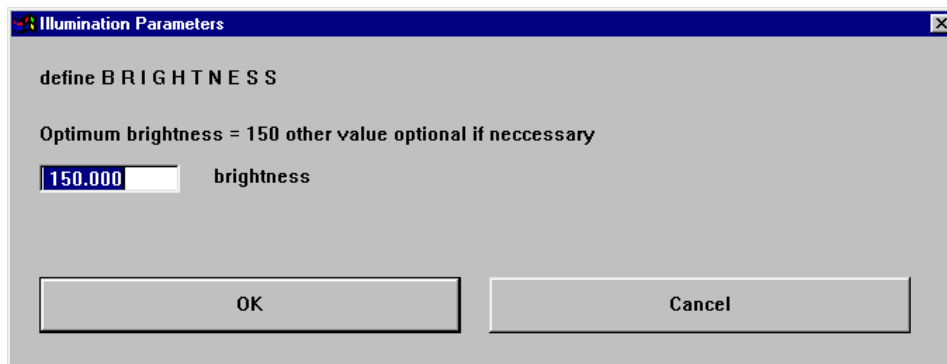
Set the aperture for each objective. Continue with "Save."

The selected apertures are valid only for reflected light.

If "Auto change" is checked they will be automatically activated when the objective is turned in.

Please, set this option for definite work of the service macros and also of the whole tool. In the Docu Window displayed in the lower right corner of the screen some recommendation for aperture setting will be given.

The last dialog box asks for the optimum brightness. By default it is 150. For all currently used cameras this is the optimum brightness.



Continue with "OK".

F5:

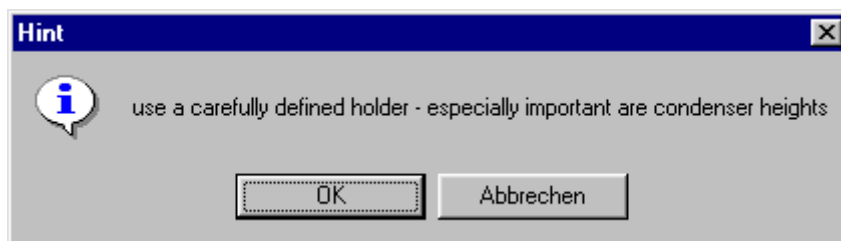
The functions F5 and F6 help exactly adjusting the condenser at tools working with transmitted light. This function supports the finding of the best condenser height. Execute it for all allowed wavelengths. The height should be checked again after the condenser is centered (F6) and also if some manipulation of the mask height in the holder had been done later (for instance during adjustment of planarity in the Planarity Macro) The height values should immediately be entered into the "Condenser..." fields of that Holder in the Holder Window the parameters of which should later be completed by learning the holder (Holder Macro). Working in transmitted light will automatically cause the condenser driving to this position – if "holder verify" was successful. The position is different for each wavelength.

F6:

This function supports the mechanically centering of the condenser. As already mentioned for F5 it should be checked again after adjustment of the holder planarity.

F7:

Helps finding the default lamp power and camera shutter values for each objective, ray path and wavelength. At first the following hint remembers that the condenser heights have to be entered into the currently active holder in the Holder Window.



Confirm this hint with OK

During a first learning – if a holder was not yet activated by executing the Holder Verify operation – the condenser will not move for changing wavelengths. Therefore F7 should be executed once again after Holder definition and activation. This is especially necessary on tools which use two or even three different wavelength in transmitted light.

Next the following box from Config Window à Microscope à Lamp Defaults appears:

Microscope Lamp Defaults TL: IR									
			Lamp Control	Power (0-255)		Shutter (ms)		fixed	
Obj. 1	2.5x IR	Reflect	☒ Power ☐ Shutter ☐ both	110.0	Set	10.00	Set	☒	
Obj. 2	5x Vis	Reflect	☐ Power ☐ Shutter ☒ both	127.0	Set	8.90	Set	☒	
Obj. 3	10x IR	Reflect	☒ Power ☐ Shutter ☐ both	113.0	Set	10.00	Set	☒	
Obj. 4	10x Vis	Reflect	☐ Power ☐ Shutter ☒ both	127.0	Set	10.70	Set	☒	
Obj. 5	63x IR	Reflect	☒ Power ☐ Shutter ☐ both	145.0	Set	10.00	Set	☒	
Obj. 6	100x Vis	Reflect	☐ Power ☐ Shutter ☒ both	163.0	Set	11.50	Set	☒	

Save Cancel

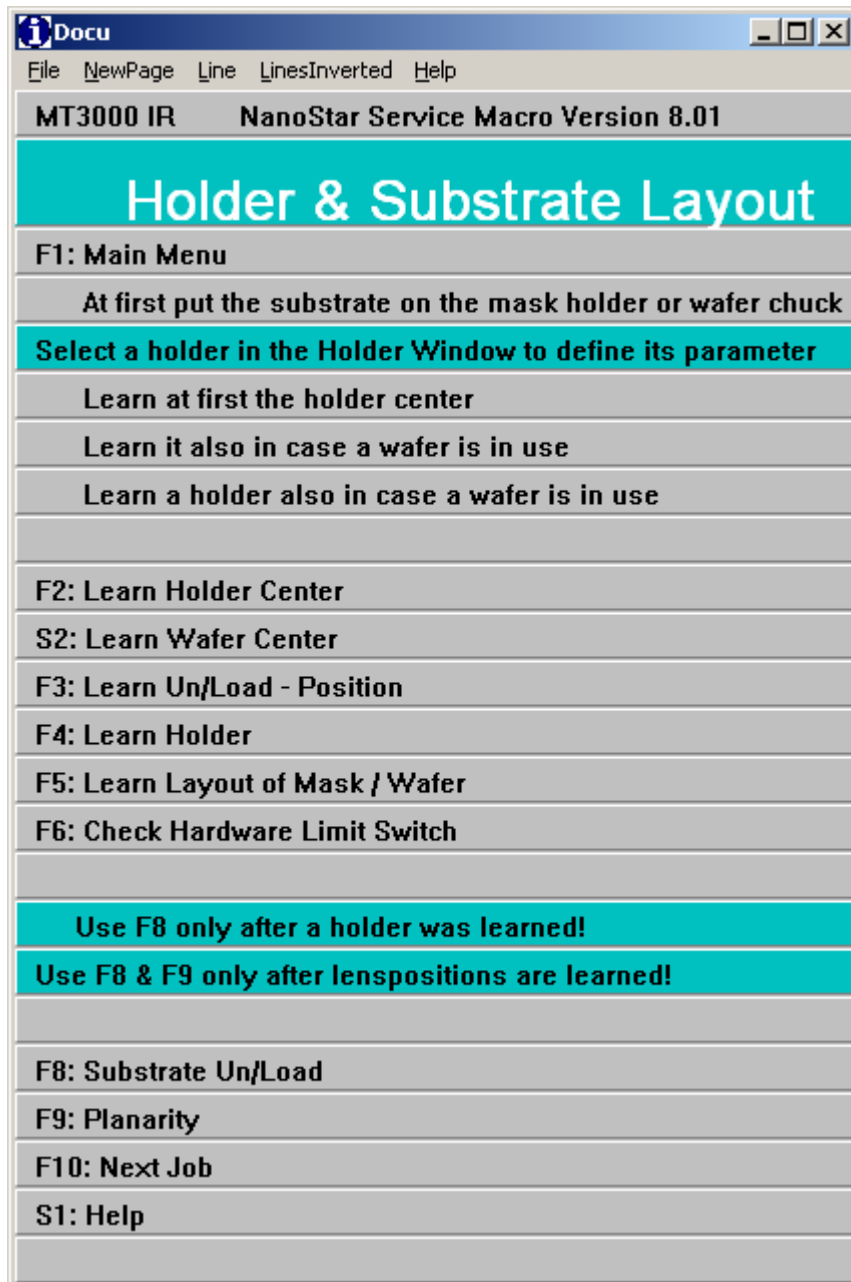
The Docu Window displays some recommendations. In a first learning cycle for visible light only the "Power" radio buttons should be set. For later use and higher speed of light-adjustment "both" can be used. This can be set when this box comes up the second time after all light adjustments are executed. Set Shutter for I-line and duv illumination, power can not vary in this case. Prefer "Power" for infrared illumination.

F8:

In difference to all other macros this function key is here not for substrate un/loading but for selecting the parameters of the test substrate - either a test mask or a test wafer.

Sample Size and Type	
Substrate Type	
☐ Mask	
☐ Wafer	
☒ Wafer for IR	
Substrate Size	
☐ 4 inch	
☐ 5 inch	
☐ 6 inch mask / 150 mm wafer	
☐ 7 inch	
☒ 8 inch mask / 200 mm wafer	
Substrate Thickness	
Mask	
☐ 9 mil	
☐ 120 mil	
☐ 250 mil	
Wafer	
☐ 680 μm = 150 mm wafer	
☒ 735 μm = 200 mm wafer	
OK	Cancel

01 Holder_and_Layout Macro



This macro helps to find out all parameters defining the substrate layout and a holder file. Learn a holder also when you work with wafers. Several important parameters are set during this routine.

F2:

F2 starts the command Holder Window → LearnCenter. The result is automatically written into the corresponding fields of the currently active holder.

S2:

S2 starts the command Stag → Alignment → Learn Wafer Center. This parameter will be used calculating some reference positions on then test wafer.

F3:

Drive the stage by joystick to the position preferred for the loading operation. The coordinates of this position are later offered to overtake them as parameters for the currently defined Holder.

F4:

The software as a guide supports learning all parameters (except condenser height) necessary to define a holder. After successfully completing this function, all parameters will be displayed in the Docu Window. They have to be edited in the Holder Window. Don't forget to "Save the edited holder" at least.

HOLDER Parameters	
Size-X = Size-Y =	8 inch
Thickness =	29 mil
Z0 =	16358.2419
Verify Objective =	4
Exchange Objective =	2
Exg-X =	6910.6049
Exg-Y =	116881.6033
condenser heights have been determined in the config macro	
Edit a new Holder in Holder Window	
Enter the Parameters and save the edited holder	
F1: Main Menu	

F5:

The software as a guide supports learning some layout specific parameters which are important for working on the test substrates in the other service macros(mask or wafer).

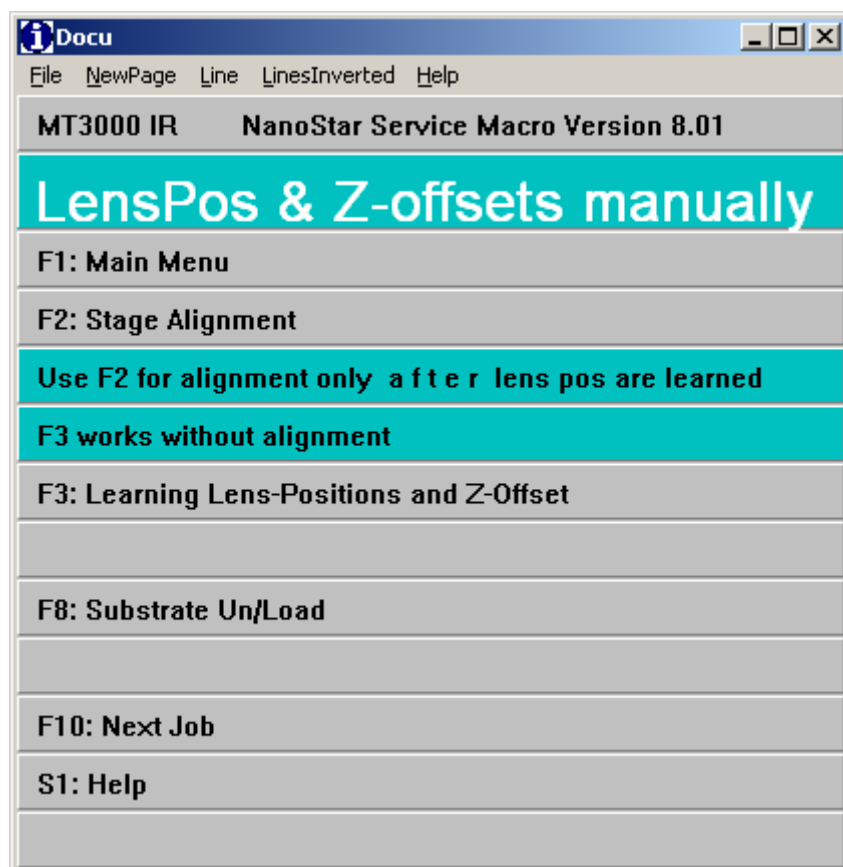
F6:

F6 executes an automatic check of correctly hardware switch setting.

F9:

F9 switches to the Planarity macro. Use it only if LensPos and z-offsets are already learned. (at least by the Lenspos_manually Macro)

02 Lenspos_Manually Macro



This macro supports the learning of lens positions and z-offsets, both parameters are important for precisely focusing with LFS and getting a nearly sharp image immediately after turning in an objective

F3:

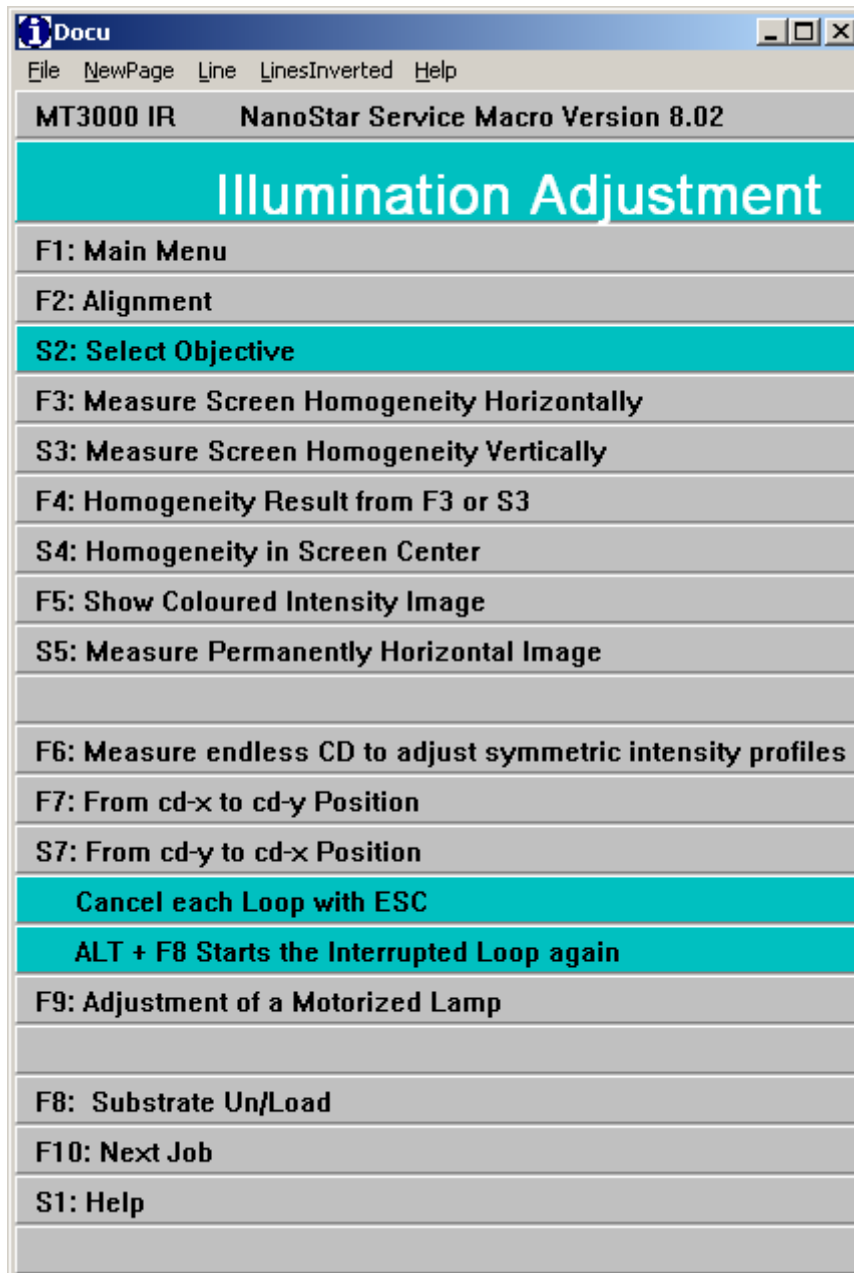
Start F3 and do what you are told by software.

The parameter LensPos is learned for the different ray paths and wavelengths. It can be learned for a selection of objectives, wavelength and path or for all possibilities.

The z-offset can be defined only if all objectives were selected, as it is the relative difference between the z positions of the objectives.

After learning is completed the values are saved into the ini file of the NanoStar software.

03 Illumination Macro



This macro supports the adjustment and evaluation of screen homogeneity.

F3: and S3:

Both functions measure the intensity in the image of an uniformly transparent or opaque area on the mask (depending on selected ray path). After the alignment (F2) is executed the positions are automatically found.

F4: and S4:

Both functions display a field of numbers, representing the relative intensity distribution in the image, resulting from the measurements with F3 or S3. S4 selects only the results of the central area in which CD measurements have to fulfill the tool performance.

F5:

The intensity distribution is displayed as a color distribution. Such a false color rendering is impressive and helpful to visually evaluate the screen homogeneity.

S6:

Starts a permanently measurement and refreshed display of results.

F6:

Supports the adjustment of illumination by displaying intensity curves at lines and spaces. The aim should be very symmetrical curves in x and y direction. Both directions are available. F6 offers at first a selection of the measurement direction.

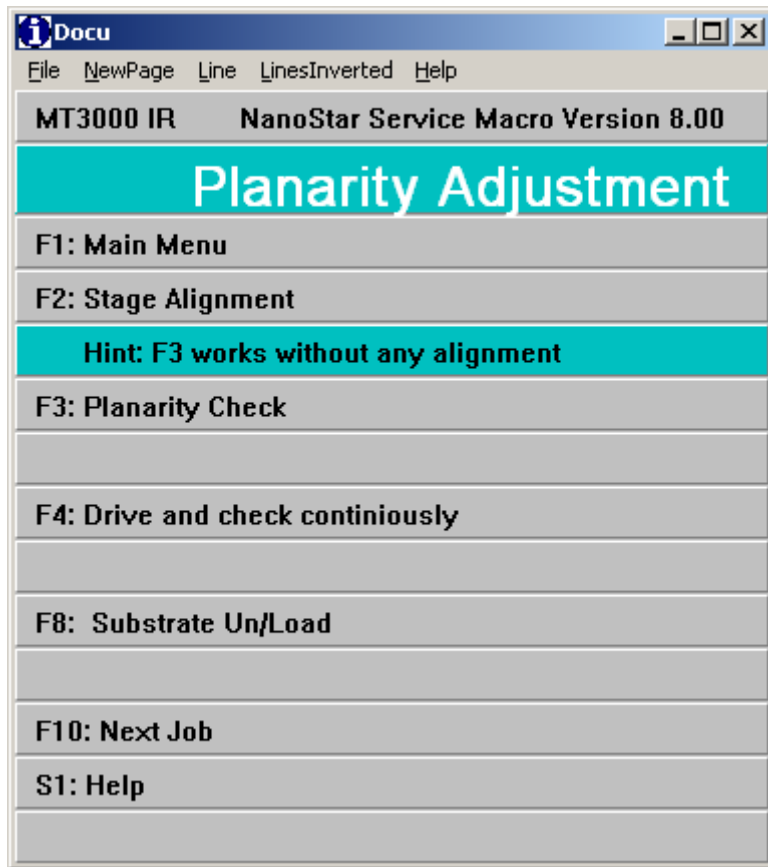
F7/ S7:

The change between the measurement position for x and y direction can be done by F7 or S7. One will not have to search for structures.

F9:

This function key starts the automatic lamp adjustment on tools equipped with a motorized lamp.

04 Planarity Macro



This macro supports the planarity evaluation and adjustment of mask holders – especially that one for the Leica test mask, used with the service macros.

F3:

The stage drives to 4 positions plus the mask center to measure there the z-position after focusing by LFS. Results are displayed as differences between the measured values of the 5 positions.

F4:

Free mode is active, LFS works, the current z height will be continuously displayed in the uppermost right corner of the screen. It's helpful to adjust the holder and getting immediately the result. A F3 check should confirm the adjustment. Do not forget to check the condenser height and probably the center position again after adjusting the holder planarity. (Configuration Macro F5 & F6). If changes of condenser parameters are necessary – learn the light defaults again (Configuration Macro F7).

05 Revolver_Offset Macro



This macro supports the adjustment of the objectives to the optical axis resp. the definition of the distances (revolver offsets) which the stage has to drive to correct any residual offset between the axis' of the objectives relative one to the other.

These stage driving-offsets are very small for well adjusted tools, except the offset to objective #1, as this one can not be mechanically adjusted. The reference objective is that one at nosepiece position five, which is fixed and can not be mechanically adjusted.

F3:

Existing revolver offsets are deleted (Stage Window à File à Revolver-Offset). This is necessary before doing the mechanical objective adjustment.

S3:

Lets the revolver rotate for mechanical fixing the objectives at their nosepiece positions. The frequently rotation can eliminate some residually mechanical play.

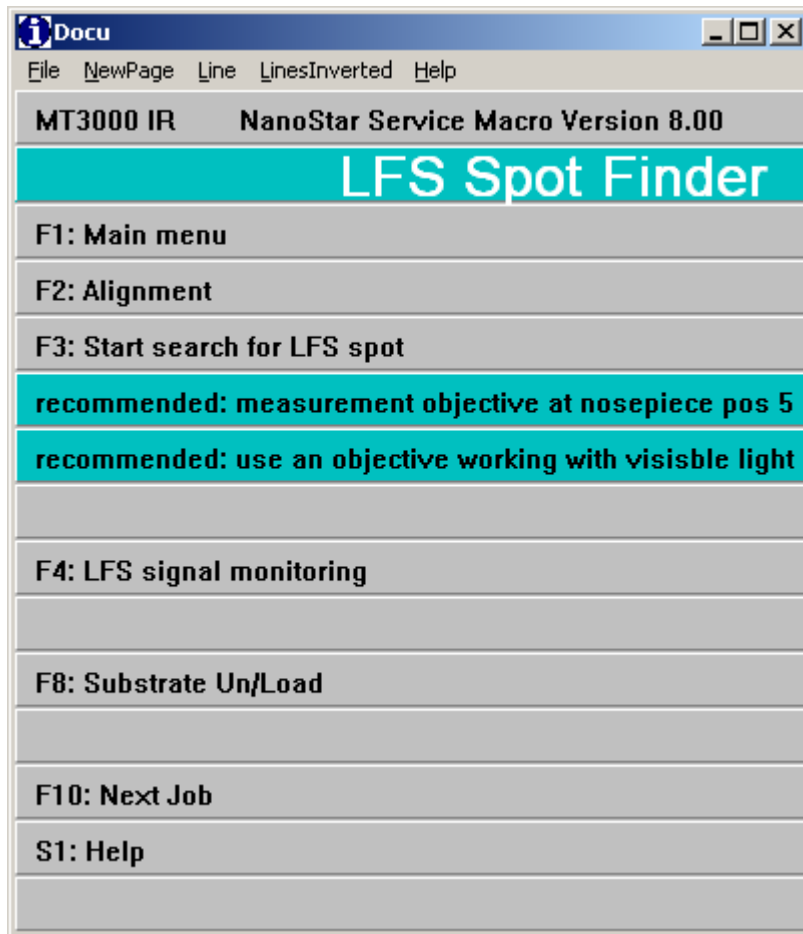
F4:

The revolver offsets of all objectives are measured and finally saved in the ini file of the NanoStar software.

F5:

Similar routine as F4 but measurements are done with the revolver offsets active. The result of this check will be displayed in a table.

06 LFSspot Macro



This macro supports the search for the position of the LFS laser beam and plots its boundaries into the IP Window. It is good to know the boundaries of the laser spot, as it should not touch any edges of structures during LFS is active. This could be a source of problems.

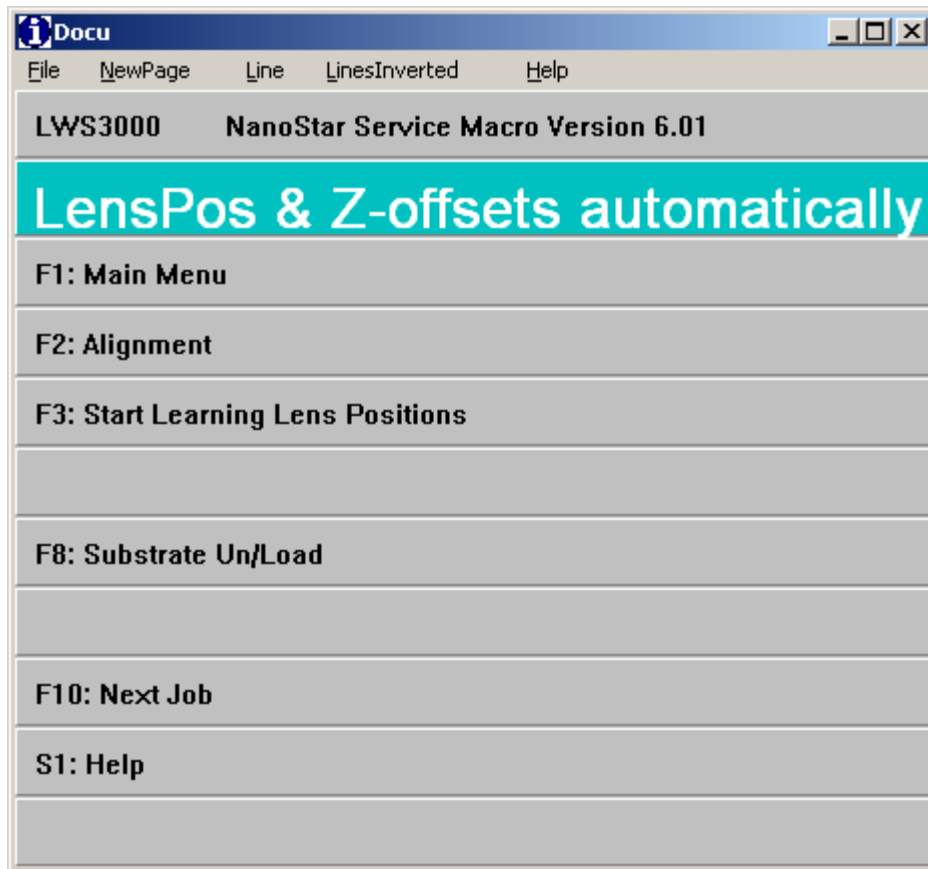
F3:

F3 starts the automatic search for the laser beam position. It is necessary to execute in advance the alignment By F2.

F4:

Monitors the laser signal in the DBgph Window as helpful tool during adjustment of the LFS laser.

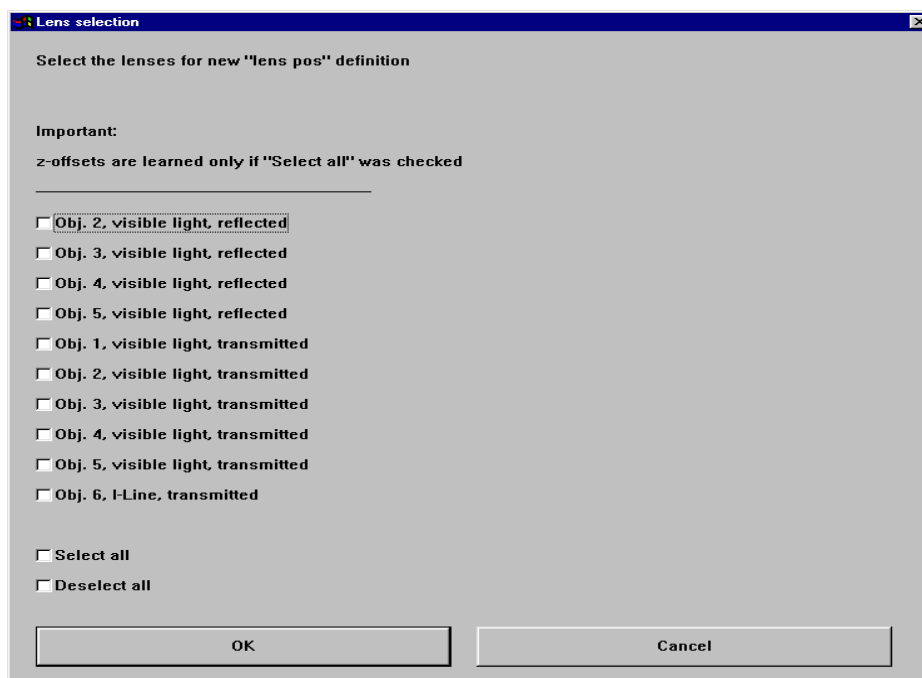
07 Lenspos_Automatically Macro



The macro supports the automatic learning of the LensPos and z-offsets for each objective and all its allowed paths and wavelengths. In difference to the Lenspos_manually macro it works independently from subjective evaluation of image focus.

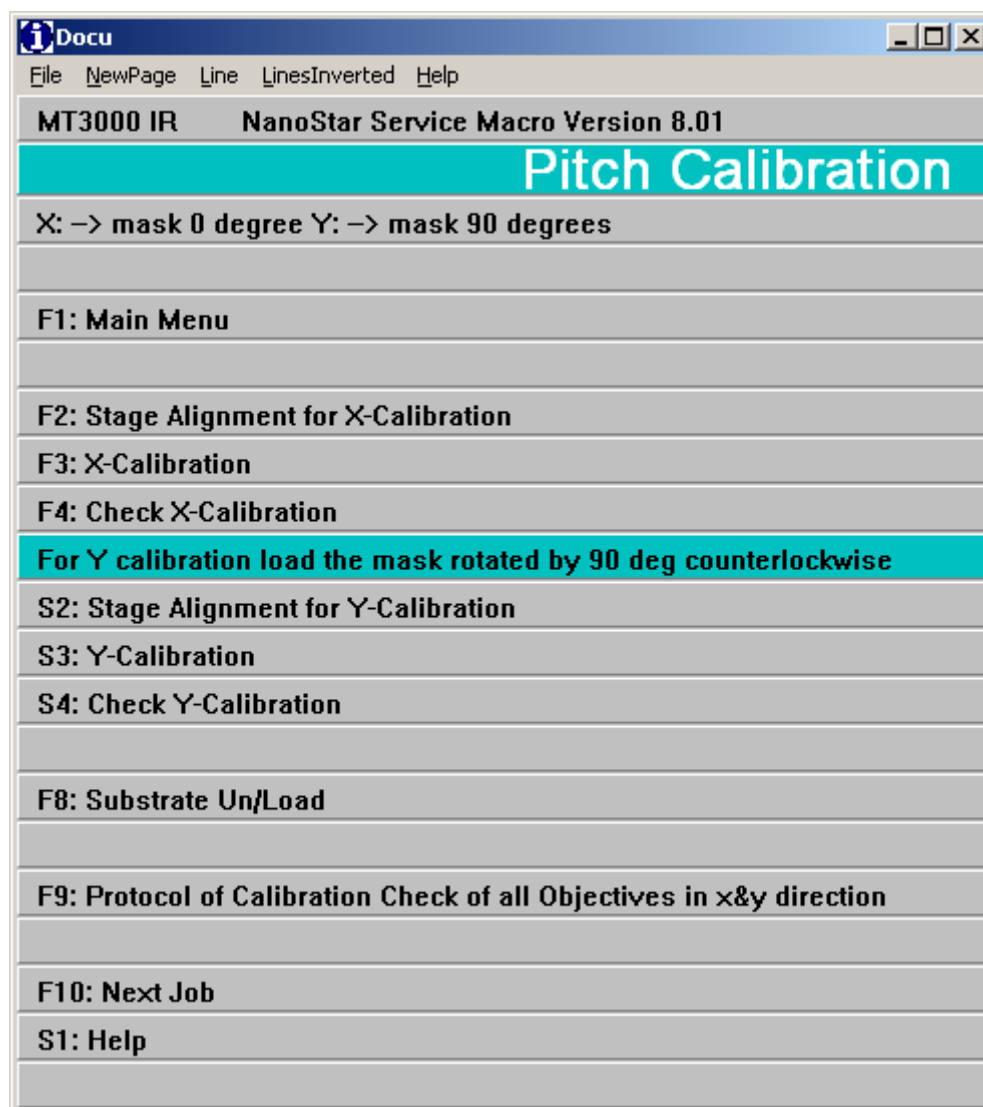
F3:

The LensPos will be automatically learned for all selected objective-path-wave combinations. The selection is done in a dialog box like the following after starting F3:



All possible combinations, defined in the config macro are offered for free choice. Important:
The z-offset can be calculated only if "Select all" is checked.

08 Pitch Calibration Macro



This macro supports the basic pitch calibration of all objectives, separately for each allowed path and wavelength.

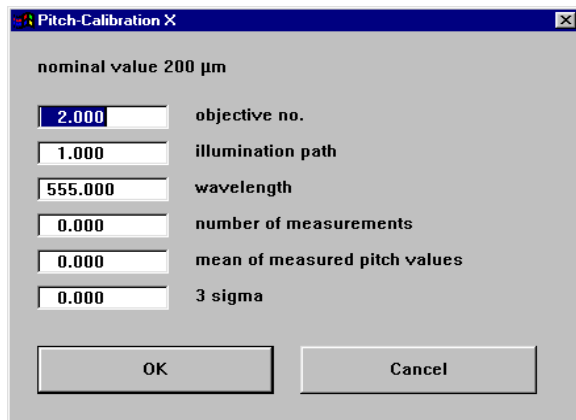
Important: use for mask loading either the F8 key or do it via Holder Window. It is important that a verify operation was successfully executed before any measurements, as the condenser height is set as defined in the Holder Window.

F2:

Alignment for measurements in x-direction.

F3:

Pitch calibration in x-direction for all selected objectives, paths and wavelengths. The dialog box for selection is similar to that one shown for the Macro "Lenspos-automatically". Additionally the number of loops must be set. By default it is 5. After all measurements are completed a box as shown below gives an overview on the measured results, for each objective-path-wavelength combination separately.



Pitch-Calibration X

nominal value 200 µm

2.000	objective no.
1.000	illumination path
555.000	wavelength
0.000	number of measurements
0.000	mean of measured pitch values
0.000	3 sigma

OK Cancel

Illumination path = 1 is reflected light, = 2 is transmitted light.

Wavelength = 555 is visible light, = 365 is I-line, =248 is DUV, =1200 is infrared.

Check carefully the results. After OK you may decide whether to use them for calibration or not. If one value is not ok – may be its 3 sigma is too large – this selection can be individually repeated after all others were accepted. After each run the command “save autaname” is executed, saving the results in a DBase file.

F4:

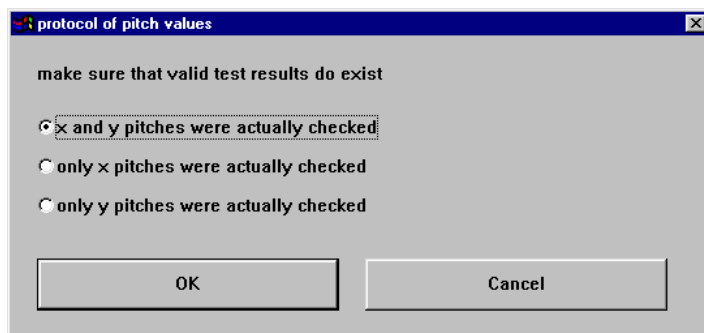
Starts nearly the same routine as F3 does, but allows only to check - not to change - the current calibration in x. The results of a check run will be saved in the checkx.db file which is used for the protocol of calibration check. Additionally after each run the command “save autaname” is executed.

S2, S3, S4

These three keys do the same as F2, F3, F4 but on the rotated mask in y-direction. Mask must be rotated by 90 °, that is the direction to make the text readable when looking from right hand side.

F9:

Displays the protocol of valid calibration checks. Decide after starting F9 which checks do exist and should be used for the protocol.



protocol of pitch values

make sure that valid test results do exist

☒ x and y pitches were actually checked

☐ only x pitches were actually checked

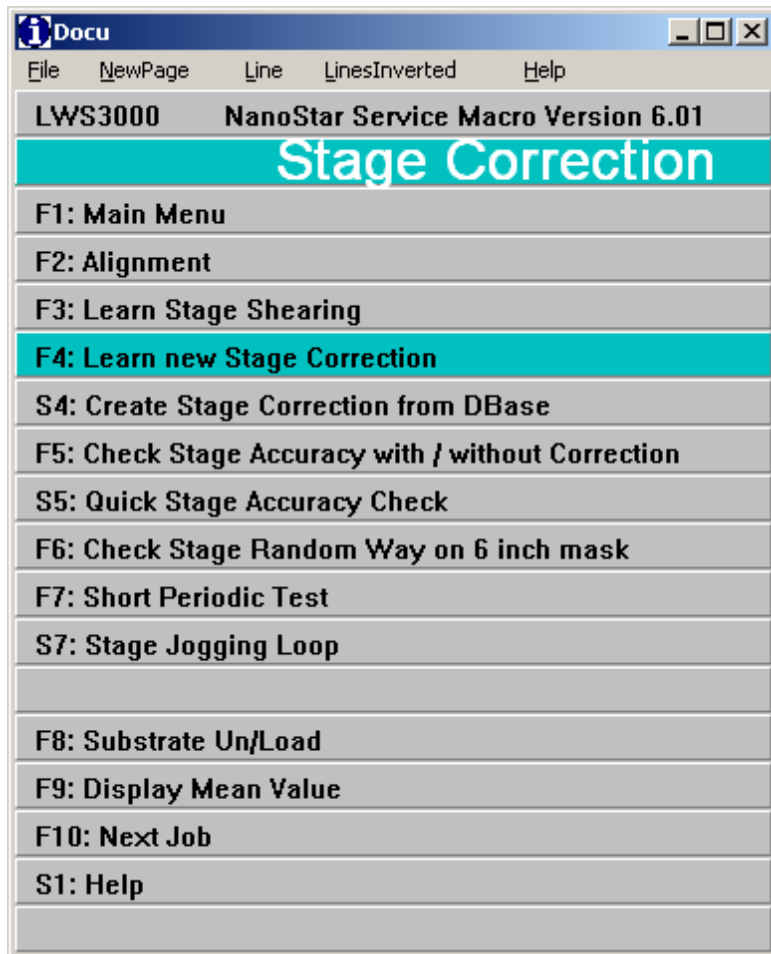
☐ only y pitches were actually checked

OK Cancel

The protocol is written for all allowed combinations of objective, path and wavelength. If no measured results do exist in the data file 0.0 will be plotted. See below an example of a protocol:

Results of Calibration Check - Pitch Values						
06.10.2005 15:52:54						
Lens Parameters		Nom.	Result X		Result Y	
		μm	Mean μm	3SD nm	Mean μm	3SD nm
visible light, reflected						
lens 2	5 x	200	200.1384	62.2	0.0000	0.0
lens 3	20 x	100	100.0874	8.8	0.0000	0.0
lens 4	50 x	16	16.0101	1.6	0.0000	0.0
lens 5	150 x	16	16.0080	0.2	0.0000	0.0
visible light, transmitted						
lens 1	2.5 x	200	0.0000	0.0	200.0255	596.9
lens 2	5 x	200	0.0000	0.0	200.0011	202.4
lens 3	20 x	100	0.0000	0.0	199.9910	5.6
lens 4	50 x	16	0.0000	0.0	99.9992	1.9
lens 5	150 x	16	0.0000	0.0	16.0002	1.6
I line, transmitted						
lens 6	150 x	200	0.0000	0.0	0.0000	0.0

9 Stage_Correction Macro



This macro supports learning and checking of all parameters which will help to drive the stage with optimum precision. All measurements in this macro are executed at special rectangle patterns, distributed all over the substrate as 8 mm grid. On the 6 inch mask it forms a 18x18 matrix, on 5 inch mask a 12x12 matrix. The layout on the wafers, of course, is not a complete matrix but uses all possible positions for such angle on the round wafer.

After starting any of these functions one has to select the substrate type and size, as well as the illumination path.

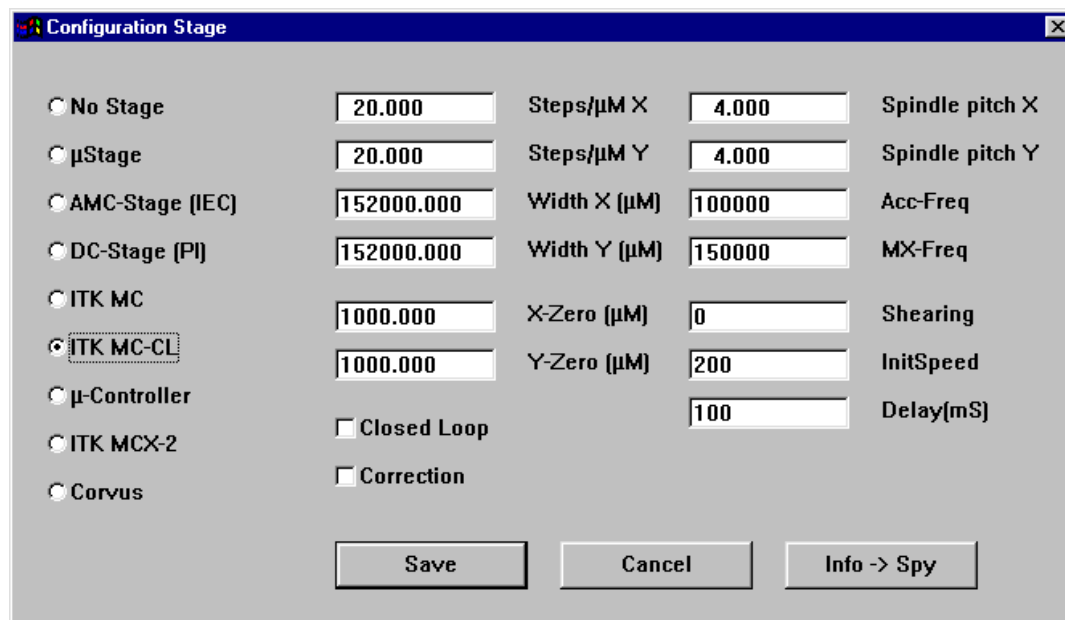
F3:

F3 measures the stage shearing parameter.

Select the objective for this measurement, and define then the substrate and stage parameters.

The boxes are the same as shown for F4.

Type the result of this run is into the entry field of the stage configuration window which appears on the screen after the run.



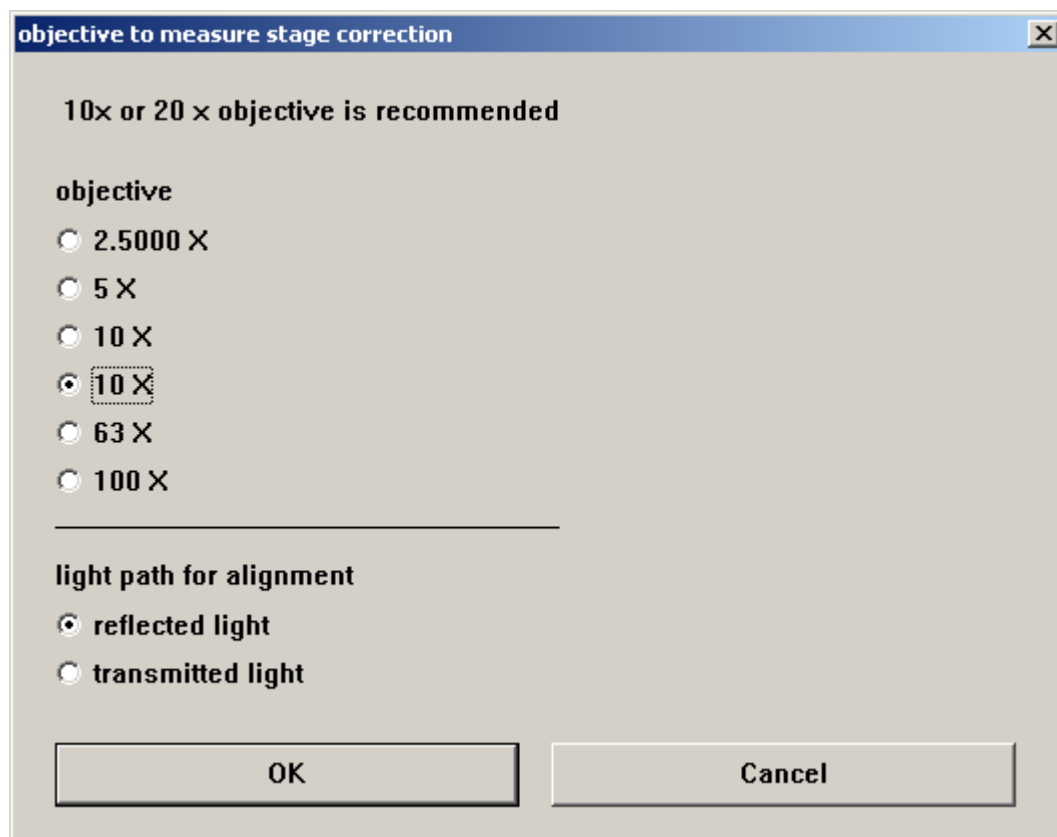
Configuration Stage

☐ No Stage	20.000	Steps/μM X	4.000	Spindle pitch X
☐ μStage	20.000	Steps/μM Y	4.000	Spindle pitch Y
☐ AMC-Stage (IEC)	152000.000	Width X (μM)	100000	Acc-Freq
☐ DC-Stage (PI)	152000.000	Width Y (μM)	150000	MX-Freq
☐ ITK MC	1000.000	X-Zero (μM)	0	Shearing
☒ ITK MC-CL	1000.000	Y-Zero (μM)	200	InitSpeed
☐ μ-Controller	☐ Closed Loop		100	Delay[mS]
☐ ITK MCX-2	☐ Correction			
☐ Corvus				

F4:

Learns the parameters for a new stage correction. To prevent errors it is seriously recommended to delete (or deactivate by changing its name) an existing stg_corr.dat file in the Nanostar directory before the first run starts. In following runs it can stay active to get more precise correction data.

Before measuring select the objective and light path. If available, transmitted light should be used as it is less sensitive to contamination like particles on the mask surface. But the macro works in reflected light as well. It is quite good – but not totally - insensitive against small defects.



objective to measure stage correction

10x or 20 x objective is recommended

objective

- ☐ 2.5000 X
- ☐ 5 X
- ☐ 10 X
- ☒ 10 X
- ☐ 63 X
- ☐ 100 X

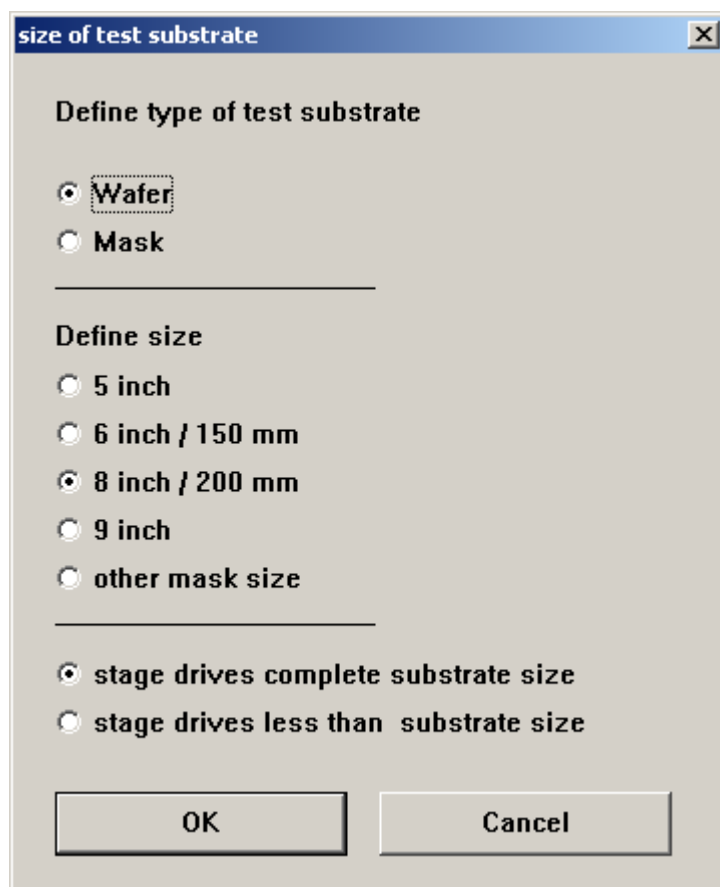
light path for alignment

- ☒ reflected light
- ☐ transmitted light

Next steps are to select the mask size and stage driving distance.

Hint:

Actually 9 inch and "other mask size" does not yet work. It's an option of the future.



size of test substrate

Define type of test substrate

☒ **Wafer**

☐ **Mask**

Define size

☐ **5 inch**

☐ **6 inch / 150 mm**

☒ **8 inch / 200 mm**

☐ **9 inch**

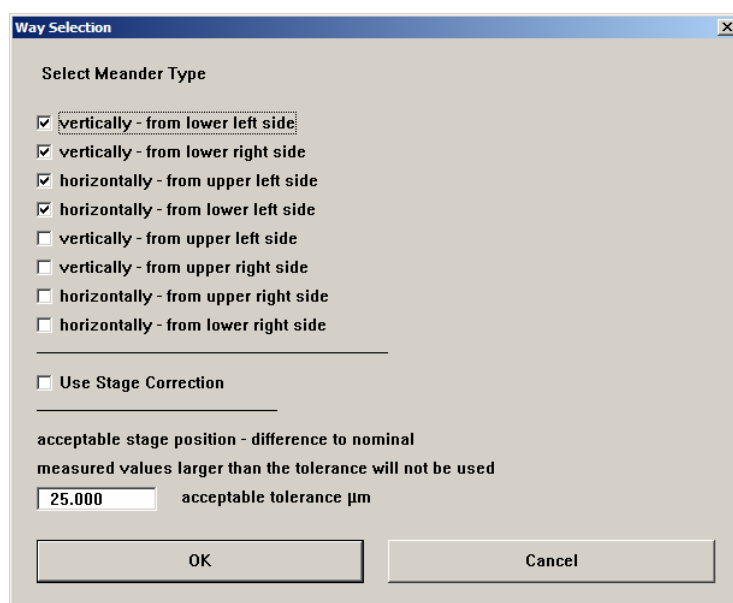
☐ **other mask size**

☒ **stage drives complete substrate size**

☐ **stage drives less than substrate size**

OK **Cancel**

The accuracy of learned parameters will be best when minimum 4 meanders were used, running the stage so that each nominal position is reached from 4 different directions. The upper 4 ways of the list in the dialog box are active by default. All 8 ways may improve the result but cost time. Check "Use stage correction" only if it is guaranteed that a valid correction data file exists, learned on a substrate of the same size as the currently used one. The selectable tolerance makes sure that values out of expected tolerance – caused by error measurements at defects or particles – will not be used for evaluation.



Way Selection

Select Meander Type

☒ **vertically - from lower left side**

☒ **vertically - from lower right side**

☒ **horizontally - from upper left side**

☒ **horizontally - from lower left side**

☐ **vertically - from upper left side**

☐ **vertically - from upper right side**

☐ **horizontally - from upper right side**

☐ **horizontally - from lower right side**

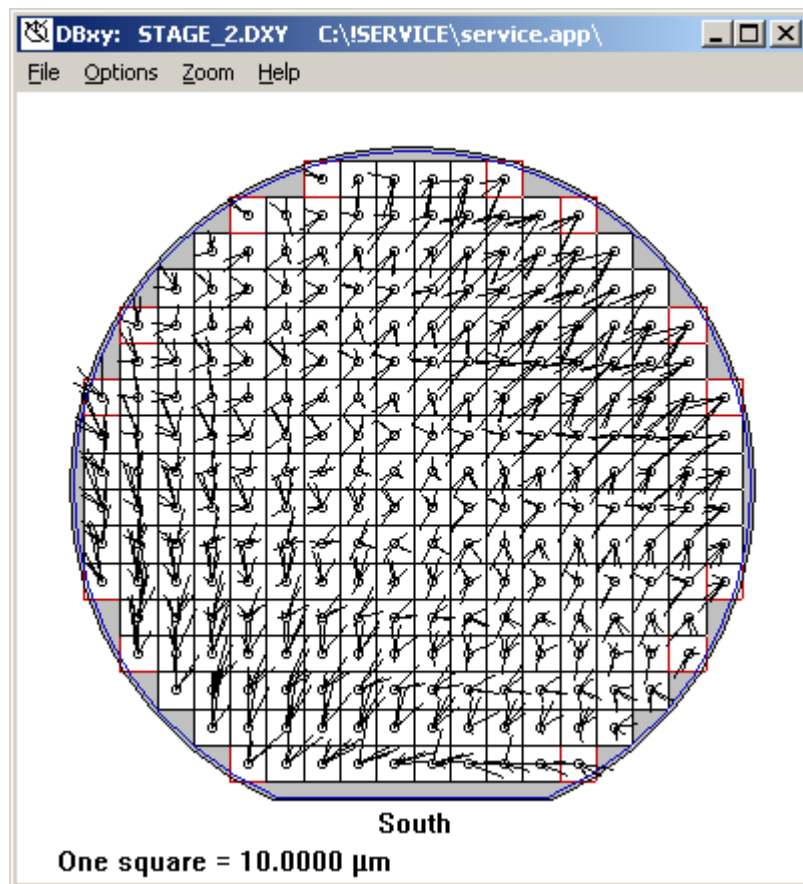
☐ **Use Stage Correction**

acceptable stage position - difference to nominal
measured values larger than the tolerance will not be used

acceptable tolerance μm

OK **Cancel**

During the run all values are plotted into the DBxy Window.



The image shows an example of a test with five loops executed on a 150 mm wafer, no stage correction active yet.

S4:

Create stage correction from DBase This function key starts the calculation of stage correction parameters using the currently active database file. As the data after each run are saved with "autoname" they can be used to reconstruct a stage correction if it was lost.

F5:

Check stage accuracy with / without correction. Widely the same procedure as F4 but it only checks the stage, creates not a new stage correction.

S5:

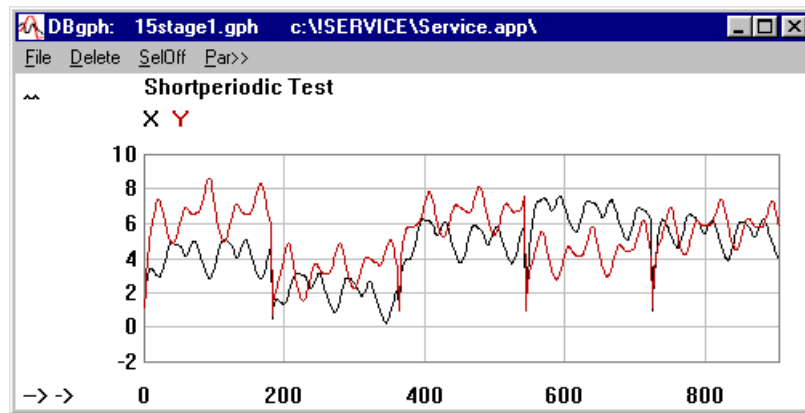
Quick stage accuracy check Checks stage accuracy but with user defined step width. Step width = 1 equals F5. With larger step width a reduced number of way positions is checked.

S6:

Check stage random way on 6 inch mask. Check of the stage accuracy similar to F5. To avoid systematic errors the positions of the 18x18 matrix are measured on a nearly stochastically driven way.

F7:

Short periodic test This is a test of stage driving spindle accuracy in a very near region. At five mask positions the stage steps horizontally and then vertically. At each screen position the difference between nominal and real position is measured. Results are displayed in the DBgph Window. Two linear plots are ideally expected at each position – but seldom reached. The results below are from a relatively bad stage, tested at all five positions.



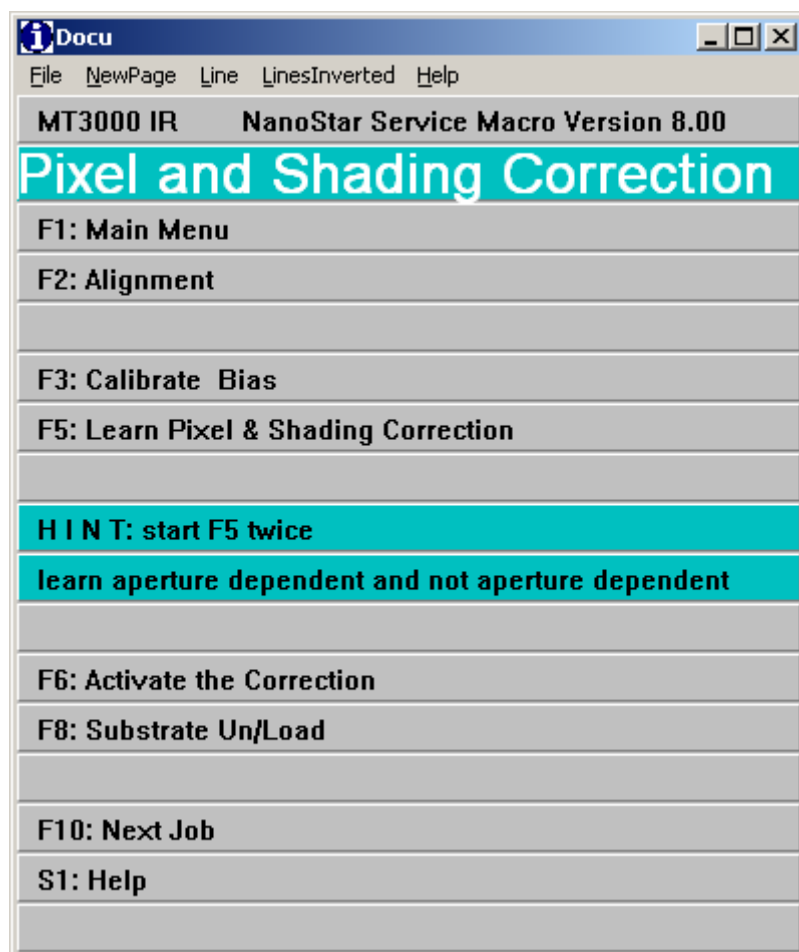
S7:

Stage jogging loop serves for stabilizing stage movement. It is the same functionality as S5, but without measuring.

F9:

Calculates the mean deviation from nominal position at each matrix point and plots it in the DBxy Window. As a consequence there will be shown only one vector in each rectangle instead of all the measured values (one to eight are possible) as shown above for F4.

10 Pixel_Shading Macro



This macro is based on the CCD Window. It supports the learning of the camera correction files. All functions are executed identically as started from the Window, but with an alignment by F2 the stage drives automatically to the positions adequate for the different calibration functions. Before the bias-, pixel-, or shading corrections are learned the joystick is free to allow for some repositioning, if any problem appears in the image field (some particle, scratch or so)

F3:

The bias is learned.

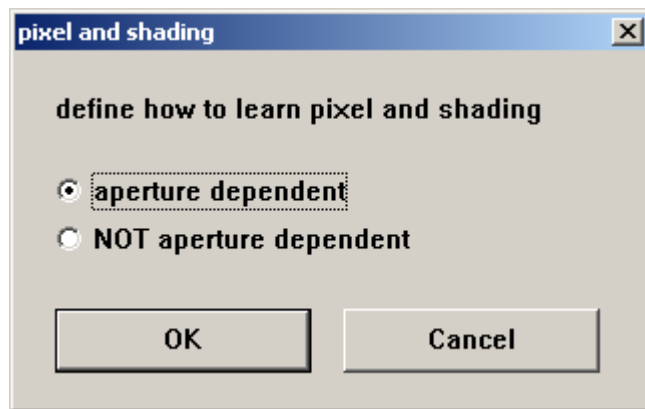
It needs to be done only once, and is then valid for all objectives and illuminations.

If the tool used more than one camera, learn it for each camera.

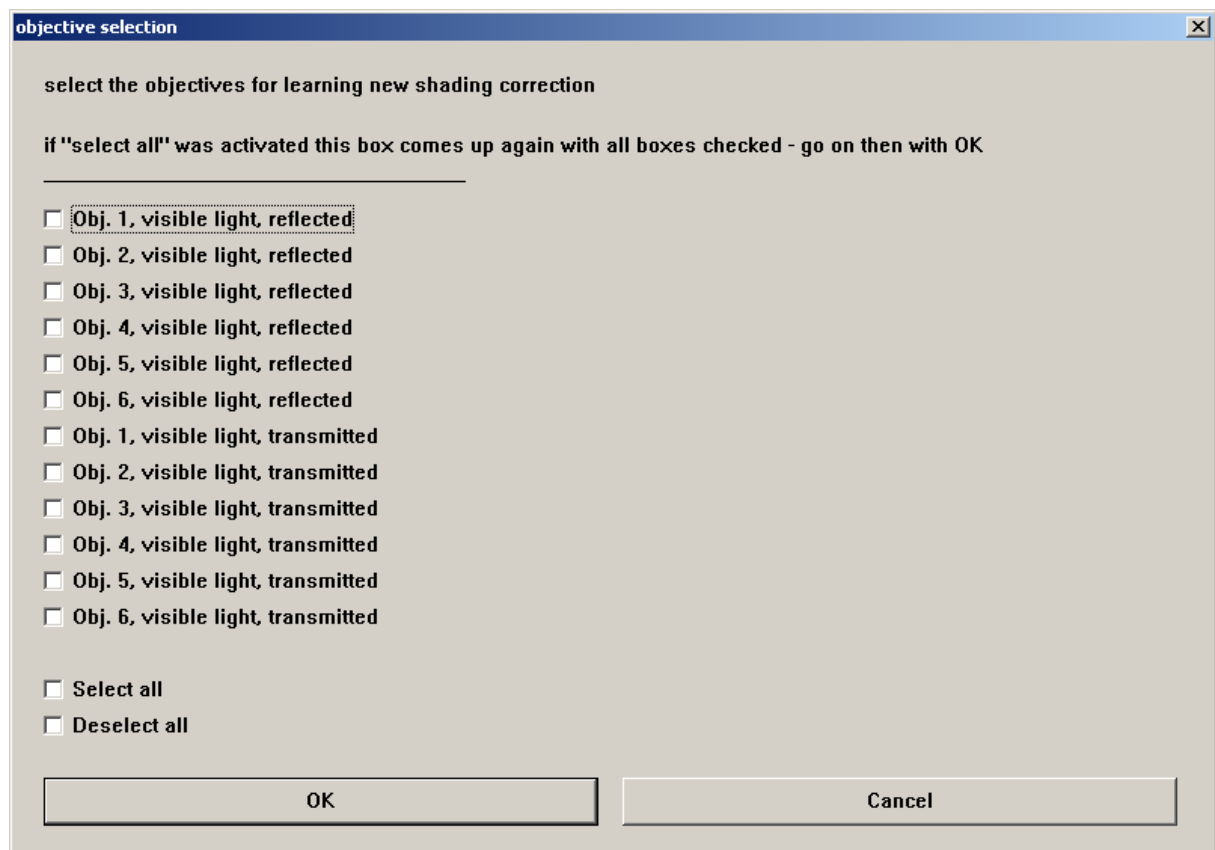
F5.

Pixel & Shading correction should be learned for each allowed combination objective-path-wavelength. Important: in reflected light you have to check, whether the aperture is set to that position which will be in use for measurements. Normally it was defined in the Configuration macro and the activated check box "auto change" in Config Window Microscope à Apertures should guaranty that it is active after revolver rotation. If auto change is not checked, aperture 3 will be set by the macro.

There is a new option to learn the corrections for each aperture. Therefore you have to decide what you want to execute.



Select then for which objectives and light paths you want to learn.



F6:

The current correction files will be automatically activated. You have to define manually for which objectives, wavelength and paths it should work.

Microscope Pixel, Shading & Distortion Correction									
			Flags			Correction Files			
			Pixel	Shading	Distortion	Pixel	Shading	Pix + Shad	Distortion
Obj. 1	2.5x IR	Reflect	☒	☒	☐	CCDpixRR	CCDsh1RR	CCDps1RR	
Obj. 2	5x Vis	Reflect	☒	☒	☐	CCDpixVR	CCDsh2VR	CCDps2VR	
Obj. 3	10x IR	Reflect	☒	☒	☐	CCDpixRR	CCDsh3RR	CCDps3RR	
Obj. 4	10x Vis	Reflect	☒	☒	☐	CCDpixVR	CCDsh4VR	CCDps4VR	
Obj. 5	63x IR	Reflect	☒	☒	☐	CCDpixRR	CCDsh5RR	CCDps5RR	
Obj. 6	100x Vis	Reflect	☒	☒	☐	CCDpixVR	CCDsh6VR	CCDps6VR	

Save Cancel

If you learned the correction depending from the aperture the upcoming box looks like the image below.

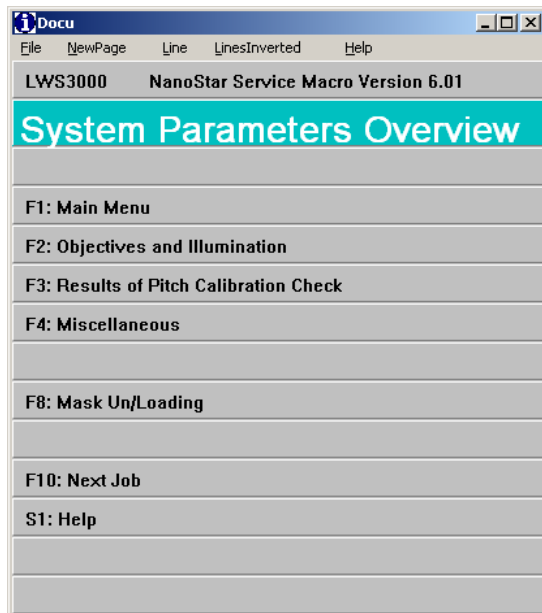
Microscope Pixel & Shading Correction TL: Vis												
			Flags		Generic	Apertures						
			Pixel	Shading								
Obj. 1	2.5x IR	Reflect	☒	☒	☒ x	☒ 1	☒ 2	☒ 3	☒ 4	☒ 5		
Obj. 2	5x Vis	Reflect	☒	☒	☒ x	☒ 1	☒ 2	☒ 3	☒ 4	☒ 5		
Obj. 3	10x IR	Reflect	☒	☒	☒ x	☒ 1	☒ 2	☒ 3	☒ 4	☒ 5		
Obj. 4	10x Vis	Reflect	☒	☒	☒ x	☒ 1	☒ 2	☒ 3	☒ 4	☒ 5		
Obj. 5	63x IR	Reflect	☒	☒	☒ x	☒ 1	☒ 2	☒ 3	☒ 4	☒ 5		
Obj. 6	100x Vis	Reflect	☒	☒	☒ x	☒ 1	☒ 2	☒ 3	☒ 4	☒ 5		

Save Cancel

Hint:

For objectives with small magnifications only apertures up to 3 are used during learning.

11 Parameter_overview Macro



This macro supports getting an overview on important parameters of the tool. A new tool installation and / or maintenance should be completed by checking all parameters and saving the protocols.

F2:

Displays all currently defined parameters for objectives and illumination.

The screenshot shows a window titled 'Draw' with a menu bar containing 'File', 'Display', 'Font', and 'Cursor'. The main content area is titled 'Lens and Illumination Parameters' and displays a table of parameters. The table has columns for 'Lense', 'LFS', 'Apert.', 'Pixel x', 'Pixel y', 'Image x', 'Image y', 'LensPos', and 'z-offset'. The data is organized into three sections: 'visible light, reflected', 'visible light, transmitted', and 'I line, transmitted'.

Lense	LFS	Apert.	Pixel x	Pixel y	Image x	Image y	LensPos	z-offset
visible light, reflected								
2 5 x	yes	1	0.6661	0.6661	852.6	682.0	13804.0	0.0
3 20 x	yes	1	0.3344	0.3343	428.1	342.4	13588.0	0.0
4 50 x	yes	1	0.0669	0.0667	85.6	68.4	13774.0	0.0
5 150x	yes	1	0.0446	0.0448	57.2	45.9	11276.0	0.0
visible light, transmitted								
1 2.5 x	yes	1	1.0000	1.0000	1280.0	1024.0	0.0	2.0
2 5 x	yes	1	0.6661	0.6661	852.6	682.0	13783.0	0.0
3 20 x	yes	1	0.3344	0.3343	428.1	342.4	13360.0	0.0
4 50 x	yes	1	0.0667	0.0668	85.4	68.4	13604.0	0.0
5 150x	yes	1	0.0446	0.0448	57.2	45.9	9808.0	0.0
I line, transmitted								
6 150x	no	1	1.0000	1.0000	1280.0	1024.0	0.0	0.0

The concrete layout of the protocol depends on the concrete tool parameters.

F3:

Displays the result of pitch calibration checks, executed in the pitch_calibration macro with F4 and S4 keys. Find an example of such a protocol at explanations for F9 in the 08 pitch_calibration Macro. It is necessary that the calibration checks were really executed. Otherwise F3 will use older data, if existing, which are not relevant for the current tool.

F4:

Displays some parameters. F4 will be subject of completing, if new parameters should additionally be displayed in future.

Tools Macros

Three macros to qualify the measurement tool belong to this set.

Acceptance_all Macro

Obj_evaluation Macro

Screenlin Macro

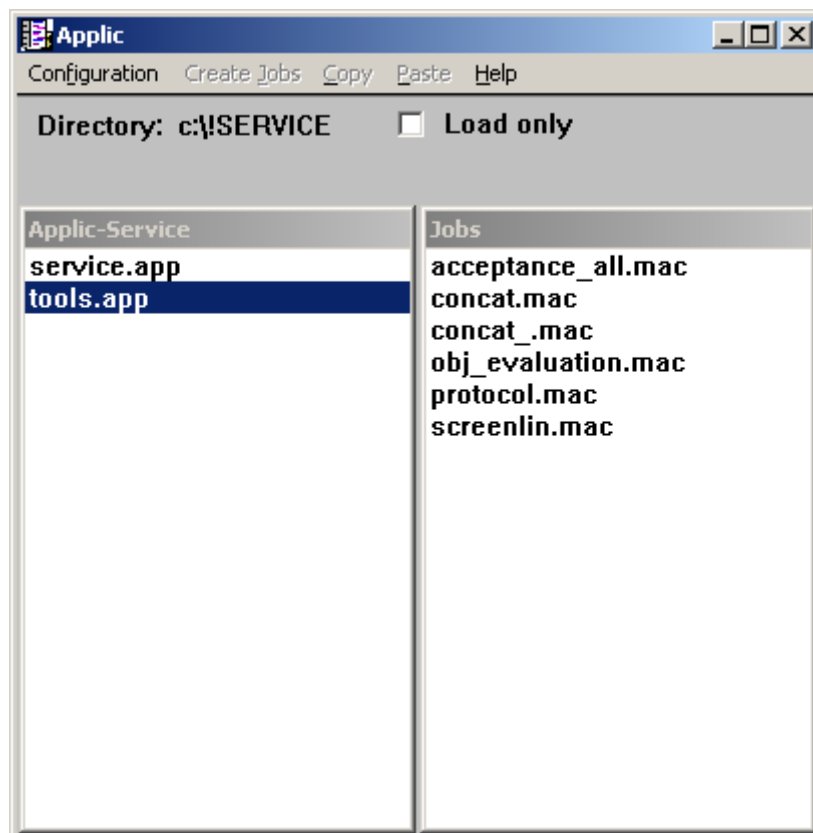
Important: starting with this version of service and acceptance macros the application directory (traditionally named as "tools.app") – containing the three macros listed above – can be stored in a freely selected path.

Acceptance Macro

The Acceptance Macro (Acceptance_all.mac) works explicitly on the MueTec Test substrate - either mask or wafer

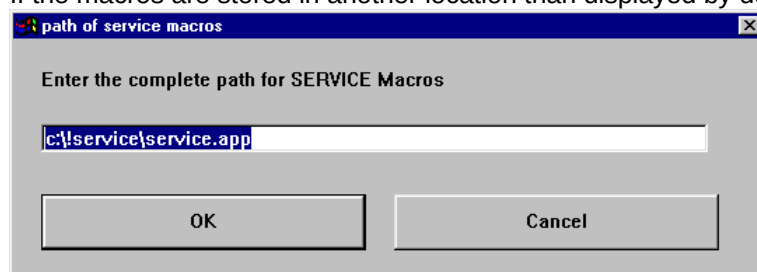
The start modus is - similar as explained for the service macros: Start à Service à Tools.app.

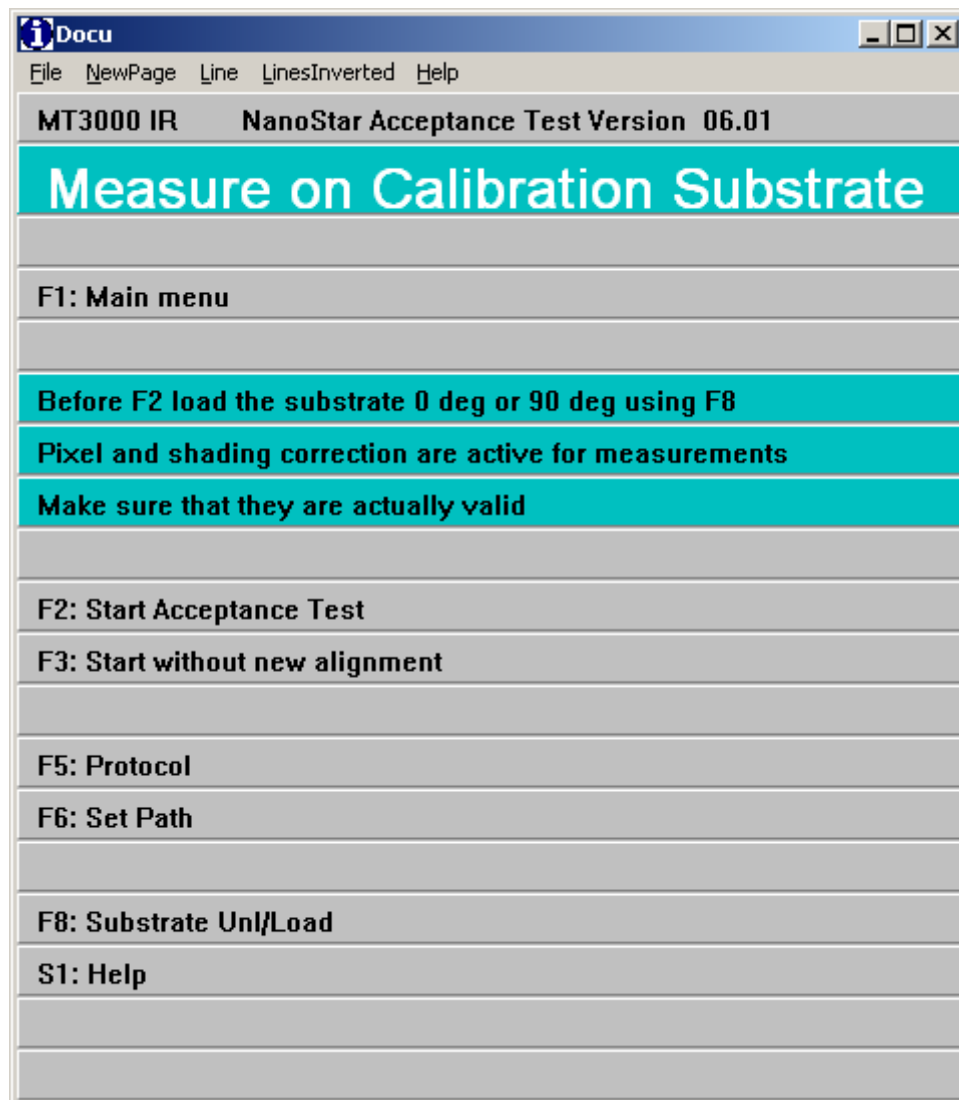
Please open for standard application the "acceptance_all.mac" The other macros are for special applications for use by MueTec service staff.



As the acceptance macro needs some files from the service macros the acceptance macro must "know" the path where those are stored. Therefore a path-check is involved. After the first start of the acceptance macro the following dialog box comes up asking for the actually valid path where the service macros are stored.

If the macros are stored in another location than displayed by default, please edit the valid name.





F2:

Starts CD measurements at lines & spaces, dots and holes, depending on the selection. The alignment is implemented into the measurement run. You have not explicitly to care for it.

F3:

Starts a further measurement, after a measurement started by F2 already was done. This saves the time for alignment.

F6:

Function to define the path containing the service macros. Use F6 after any changes of storage position of the service macros – not the acceptance macro.

F9:

The protocol of all measured results – in x and y direction - will be displayed. The illumination wavelength, ray path, structure type will automatically be recognized.

Starting F2 the following selections are requested:

Measurement Conditions [X]

set job parameters

Illumination wavelength

☐ visible

☐ i-line

☐ duv

☐ IR

Illumination Path

☐ Transmitted

☐ Reflected

Nosepiece Position of Measurement Objective

nosepiece pos of measurement objective

Selection of CD-Nominal to be measured

☐ CD for acceptance test

☐ measure all

☐ CD free selected

Select structure type for measurement

all selection is referred to measuring lines and holes in the mask center
dots then are measured at the structure group at the right side of the center

☐ lines and spaces

☐ dots - use not with IR illumination

☐ holes - use not with IR illumination

OK **Cancel**

Important: Do not measure dots and holes if measurements are executed at one of the excentric groups (as you will decide some what later). The stage coordinates used in the macro are not correct then.

Selection of CD-Nominal:

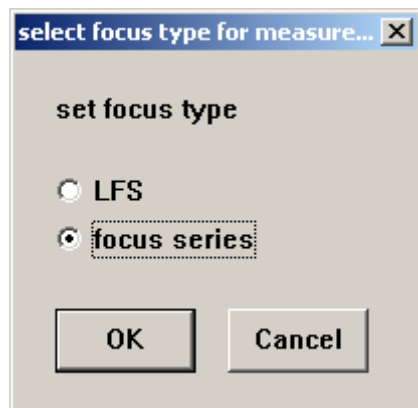
CD for acceptance test will measure a defined group of structure sizes. The sizes depend on the wavelength only not on the magnification of the measurement objective. It is assumed that 100 or 150x magnification is used for acceptance tests or 63x with infrared illumination. For smaller magnification, please select freely the structures to measure. In visible light 0.5; 1.1; 2.2; 4.6; 8.0 μm will be measured; in I-line additionally 0.3 μm are measured; DUV takes the following sizes 0.3; 0.5; 0.7 0.9 1.1; 2.2 μm . Infrared will measure 2.2; 4.6; 8.0 μm

"Measure all" for visible and I-line takes all the following sizes, DUV will stop with 4.6 μm .

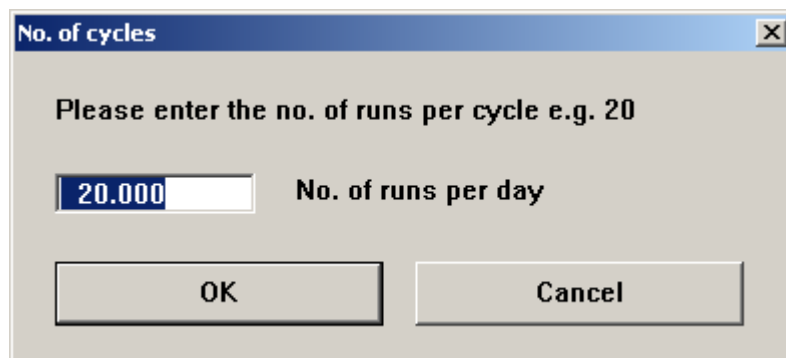
0.3; 0.4; 0.5; 0.6; 0.7; 0.8; 0.9; 1.1; 1.3; 1.5; 1.8; 2.2; 2.6; 3.2; 3.8; 4.6; 5.4; 6.5; 8.0 μm .

Dots and holes exist for all nominal values up to 4.6 μm .

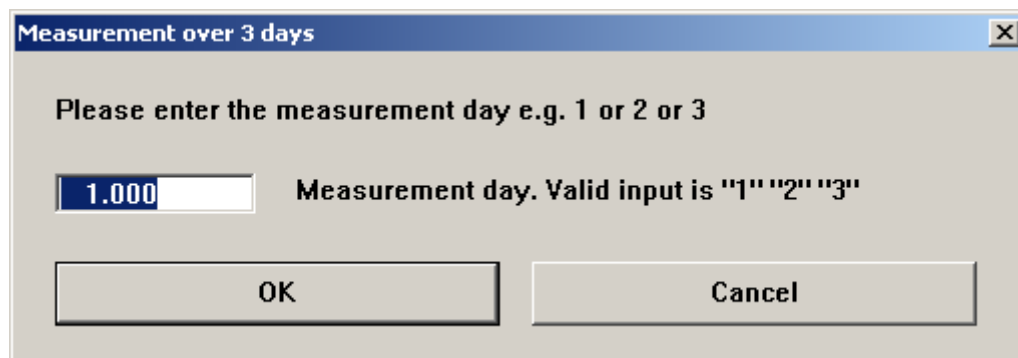
Than the type of focusing should be selected.



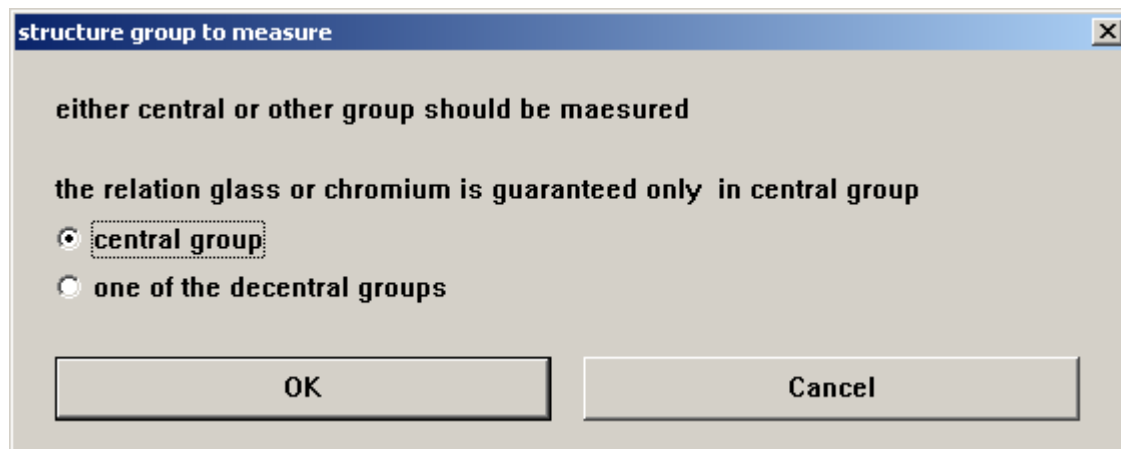
Next decision is to define the number of measurement loops.



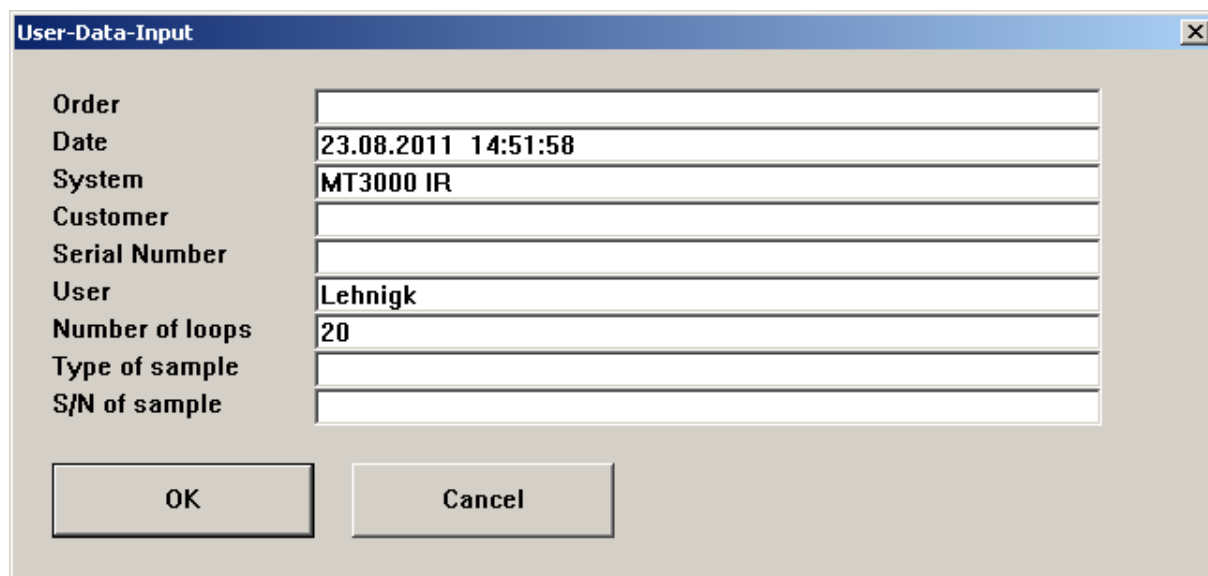
Select then the "day". The day is used to separate data for the "three days acceptance test" and also for evaluating the results in the protocol. Additionally to saving results as "day" files after each completed run a "save autname" is executed. Results from x or y directions are saved in differently named files: day_1.db resp. day_1_90.db



Decide now whether to measure at the structure group in the center of the mask or at another one.



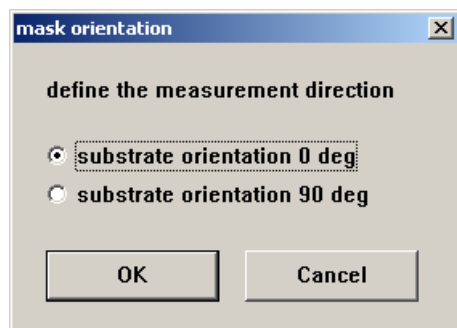
Next window coming up is from User Window allowing to enter some data to save with the database file.



A dialog box titled "User-Data-Input" with a close button (X) in the top right corner. It contains a list of labels on the left and corresponding text input fields on the right. The labels are: Order, Date, System, Customer, Serial Number, User, Number of loops, Type of sample, and S/N of sample. The input fields contain the following values: Date is "23.08.2011 14:51:58", System is "MT3000 IR", User is "Lehnigk", and Number of loops is "20". The other fields are empty. At the bottom, there are two buttons: "OK" and "Cancel".

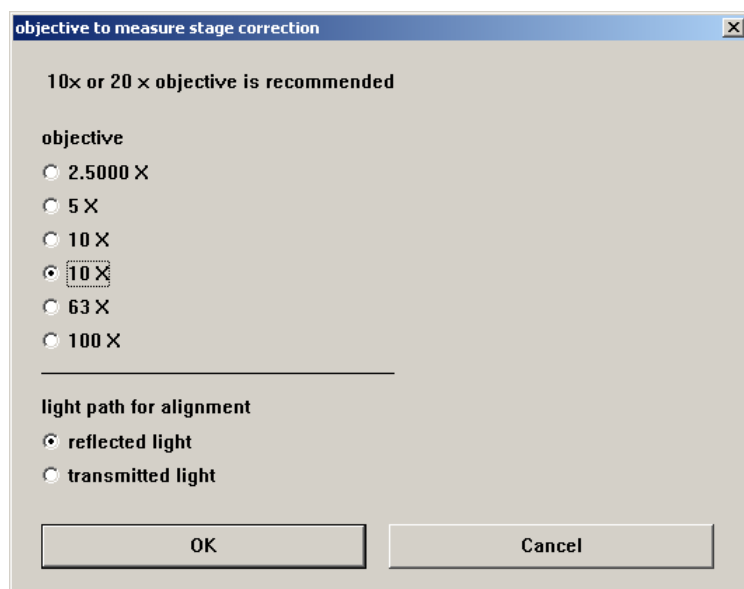
Label	Value
Order	
Date	23.08.2011 14:51:58
System	MT3000 IR
Customer	
Serial Number	
User	Lehnigk
Number of loops	20
Type of sample	
S/N of sample	

Next question refers to substrate orientation.



A dialog box titled "mask orientation" with a close button (X) in the top right corner. It contains the text "define the measurement direction" followed by two radio button options: "substrate orientation 0 deg" (which is selected) and "substrate orientation 90 deg". At the bottom, there are two buttons: "OK" and "Cancel".

Select now the objective and light path for stage alignment

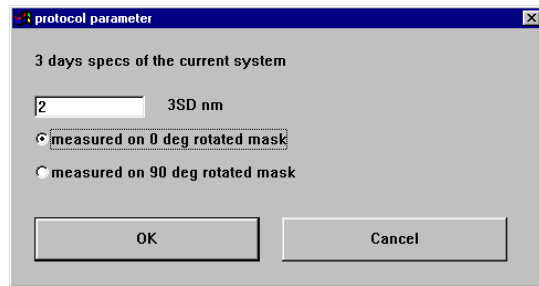


A dialog box titled "objective to measure stage correction" with a close button (X) in the top right corner. It contains the text "10x or 20 x objective is recommended" followed by a section labeled "objective" with seven radio button options: "2.5000 X", "5 X", "10 X", "10 X" (which is selected), "63 X", and "100 X". Below this is a section labeled "light path for alignment" with two radio button options: "reflected light" (which is selected) and "transmitted light". At the bottom, there are two buttons: "OK" and "Cancel".

The job will run now until data are saved. The "day" is used to separate data for the "three days acceptance test" and also for evaluating the results in the protocol. Additionally to saving results as "day" files after each completed run a "save autoname" is executed. Results from x or y directions are saved in differently named files: day_1.db resp. day_1_90.db

F9:

Creates the protocol of 3 days test runs. One has to select for which measurement direction the protocol should be generated. Additionally one has to enter which 3SD specs (nm) are valid on the current tool.



Please, make sure, that currently valid results from “3days” which really belong together –do exist. Otherwise the 3SD values can not be used.

The protocol looks like the example below. Its actual layout and content will depend on what was really measured.

Draw File Display Font Cursor

Protocol printed on: 11.10.2005 15:28:04

ACCEPTANCE TEST PROTOCOL visible, transmitted

Test-Mask Orientation 90 deg lines & spaces

ORDER : test SERIAL NUMBER :
 CHECK-DATE : 27.09.2005 15:54:28 TESTED BY : Birkenmayer
 SYSTEM : LWS3000 REPETITIONS : 1
 CUSTOMER : TYPE OF SAMPLE :
 S/N OF SAMPLE :

GLASS LINES results in micrometers

Nominal	1.DAY		2.DAY		3.DAY		OVER ALL		SPECS
	mean	3 SD	mean	3 SD	mean	3 SD	mean	3 SD	3 SD
0.5	0.5026	0.0000	0.5020	0.5464	0.5020	0.5464	0.5021	0.4999	0.002
1.1	1.0963	0.0000	1.1029	0.4874	1.1029	0.4874	1.1019	0.5005	0.002
2.2	2.2033	0.0000	2.2056	0.3632	2.2056	0.3632	2.2053	0.4902	0.002
4.6	4.6001	0.0000	4.6100	0.3184	4.6100	0.3184	4.6086	0.5188	0.002
8.0	8.0097	0.0000	8.0127	0.3577	8.0127	0.3577	8.0122	0.5468	0.002

GLASS LINES results in micrometers

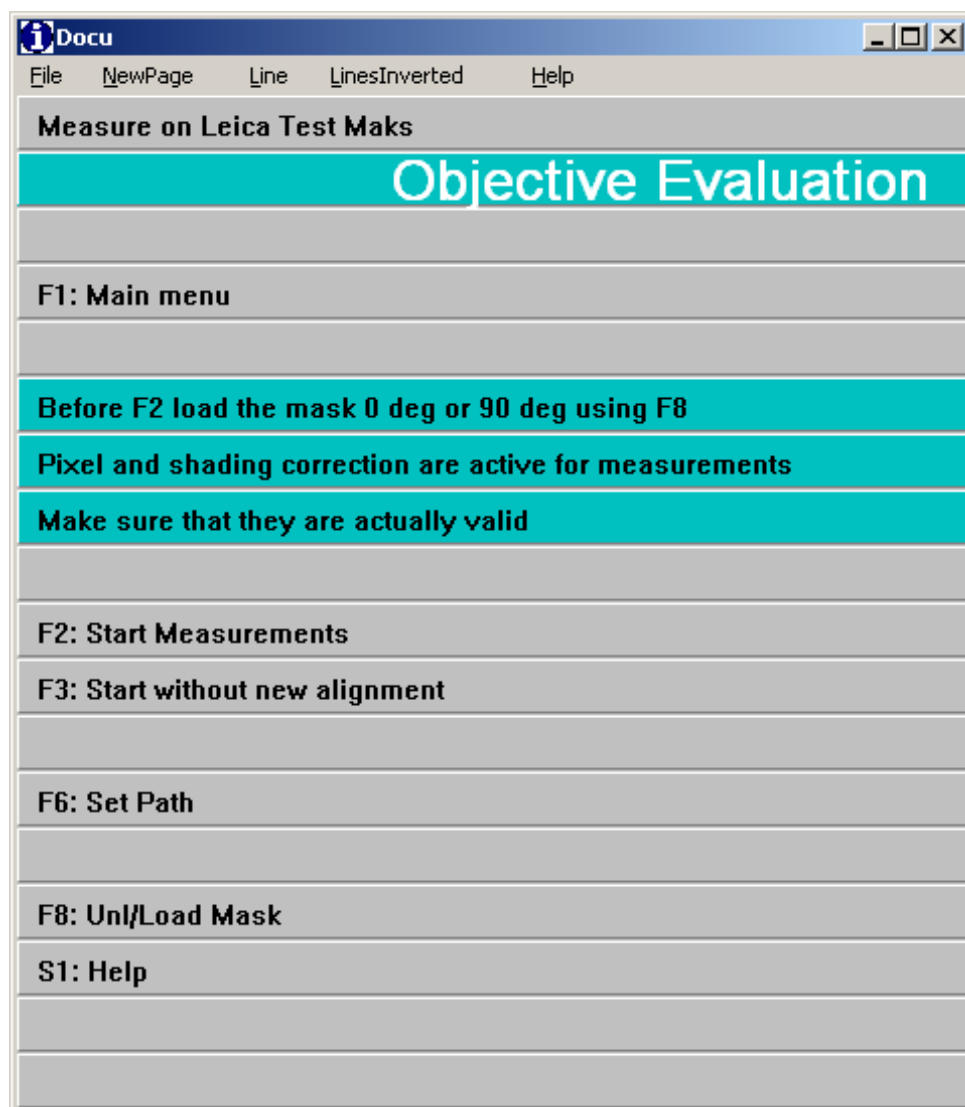
Nominal	1.DAY		2.DAY		3.DAY		OVER ALL		SPECS
	mean	3 SD	mean	3 SD	mean	3 SD	mean	3 SD	3 SD
0.5	0.5026	0.0000	0.5020	0.5464	0.5020	0.5464	0.5021	0.4999	0.002
1.1	1.0963	0.0000	1.1029	0.4874	1.1029	0.4874	1.1019	0.5005	0.002
2.2	2.2033	0.0000	2.2056	0.3632	2.2056	0.3632	2.2053	0.4902	0.002
4.6	4.6001	0.0000	4.6100	0.3184	4.6100	0.3184	4.6086	0.5188	0.002
8.0	8.0097	0.0000	8.0127	0.3577	8.0127	0.3577	8.0122	0.5468	0.002

CONFIRMED BY SIGN:

Different protocol sheets are produced for lines and spaces, dots, holes, one after the other. After one section is calculated one can decide, whether to print or save as .pcx the current sheet of the Draw Window.

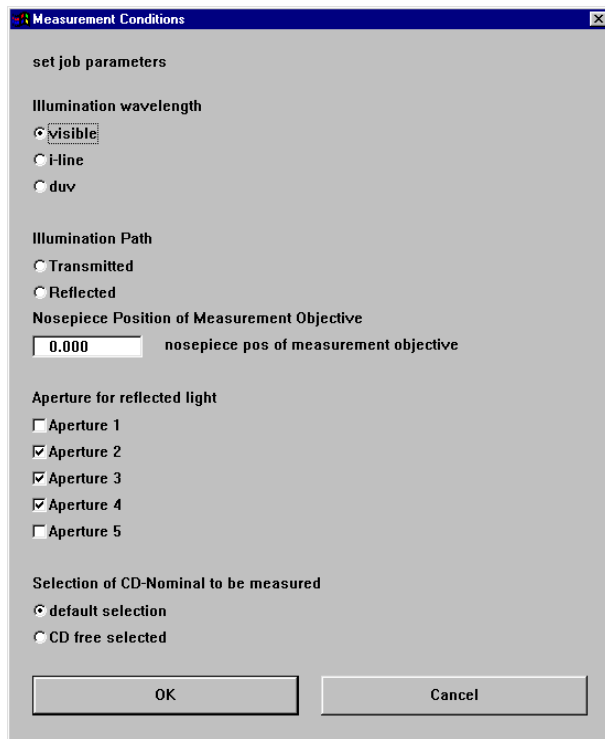
For looking at the results without saving them, please click "CANCEL" and drag the Draw Window to see all values you are interested in. Go on with "CTRL + F8". This hint is also given in the Docu Window which appears on the screen simultaneously with the Draw Window.

Obj_evaluation Macro



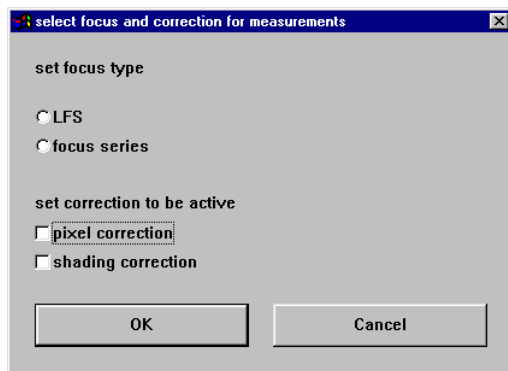
Important: this macro is not yet adapted to MT3000IR tools. (August 2011)

This macro can be used to measure the resolution of objectives using the structures on the Leica Testmask. It is set upon the same basic routine as the acceptance macro is.

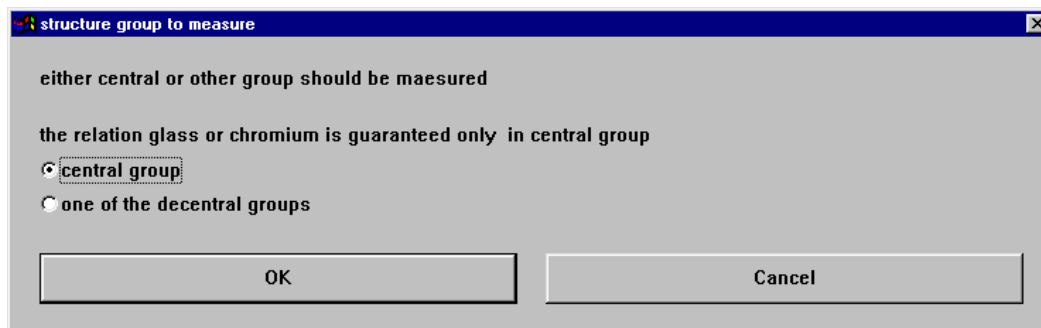


Parameter selection is somewhat different to the acceptance macro. Besides wavelength, path, and nosepiece position the apertures for reflected light and the cd nominal of measured lines and spaces must be defined. In most cases the parameters will be used as they are offered preselected (see the dialog box above).

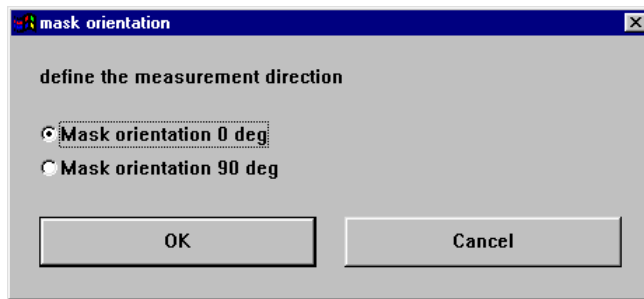
Next set the type of focus and whether the camera corrections should be active or not. (This is no subject of selection in the acceptance macro.)



Then decide where to measure. It is recommended to use the central group, because the comfortable evaluation of the measured results in a special EXCEL program relates to chromium / glass as existing in mask center.



Last but not least the mask orientation must be told to the software.



To have all important information saved in the DBase File together with the results a relatively large user request must be filled in.

Please do it carefully.

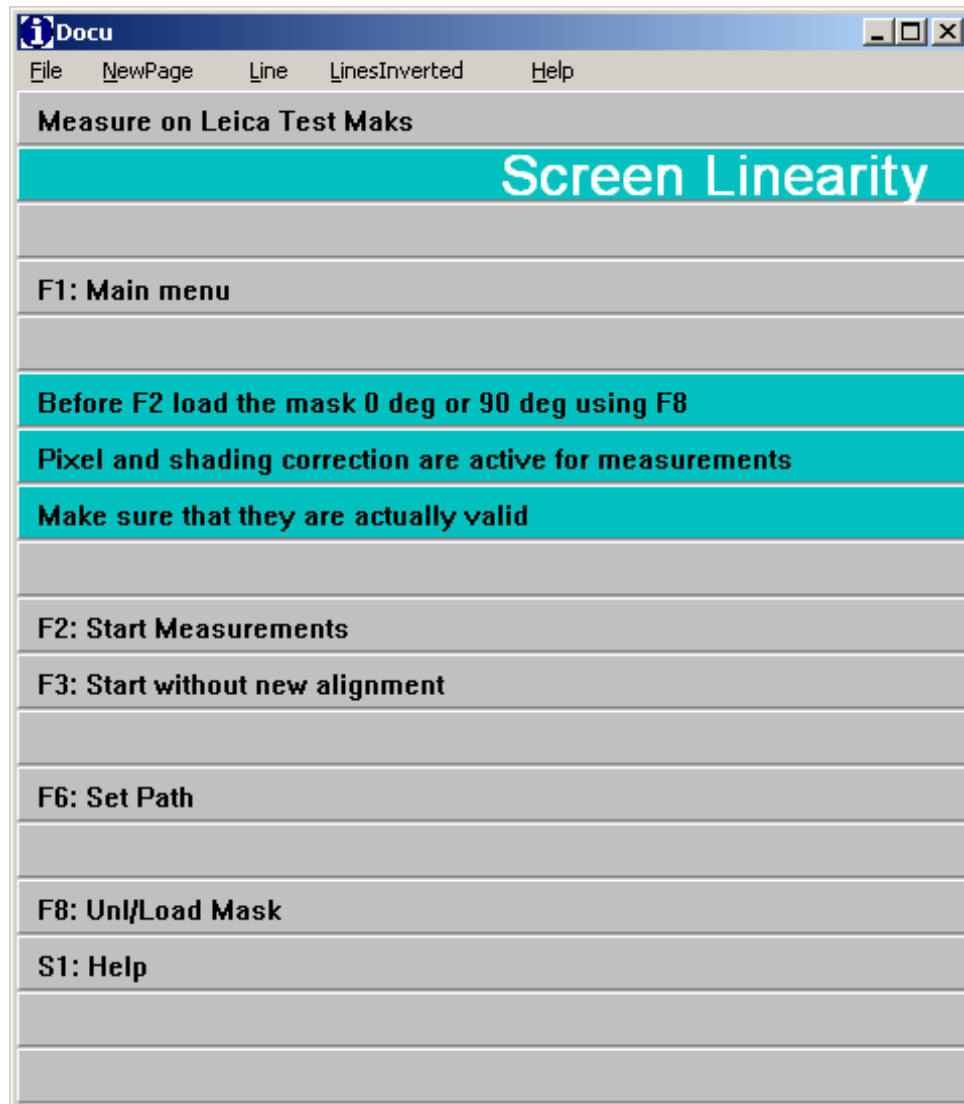
The inputs are part of evaluation in EXCEL.

As the macro was written for use by MueTec staff many of the text lines are currently in German language.

Datum	10.11.2006 15:08:25
Standort	München
Gerätesystem u. Tool #	
Mikroskop Typ u. Nr.	INM200 000123/BZ06
Objektiv Typ	HCX PL APO BD
Objektiv Charge u. SN	H5GH/00 mue 00116
Objektiv Vergr. u. NA	100x 0.90
Revolverposition	0
Tubuslinse	
Kamera Typ u. SN	JAI CV-M4+CL SN:E4 33903D
Lichtpfad	transmitted
Lichtart	visible
Maske	Leica 1/3/9484-CM01-01
Kommentar	WS3000 in München
Art der Messung	Resolution Mask 0 deg

The macro will run automatically and save a data file for each programmed combination of objective – path – wavelength – aperture. The file names are self-explaining.

Screenlin Macro



Important: this macro is not yet adapted to MT3000IR tools. (August 2011)

After start via F2 or F3 the decisions asked in the dialog box shown below must be done. Additionally to the obj_evaluation macro one has to select the size of the contacts which are measured in x and y to evaluate the screen linearity. Whether to measure dots and holes or only one type of them must be decided as well as the size of the check-matrix. Best is to follow the recommendation.

All further questions are identical to the obj_evaluation macro and are not explained again.

Measurement Conditions

set job parameters

Illumination wavelength

☒ visible

☐ i-line

☐ duv

Illumination Path

☐ Transmitted

☐ Reflected

Nosepiece Position of Measurement Objective

5 nosepiece pos of measurement objective

Aperture for reflected light

☐ Aperture 1

☒ Aperture 2

☒ Aperture 3

☒ Aperture 4

☐ Aperture 5

Selection of CD-Nominal to be measured

☒ 2.2 µm for high magnification

☐ 4.6 µm for 50x magnification

☐ CD free selected

Select structure type for measurement

☒ dots

☒ holes

Define the measurement matrix - minimum an array of 3x3

5 number of x positions

5 number of y positions

OK Cancel

To get a standardized overview on the results of the measured results there exist three EXCEL programs:

toolmatching_check_vorlage, toolmatching overview_vorlage, and toolmatching_macros

These excel programs are nearly self explaining.

Open toolmatchgng:macros in any case. It has to run in the background.

Use toolmatching_check to display the results separately for each objective and each aperture.

Both results from obj_evaluation and from screenlin are necessary to fulfill such a sheet.

Toolmatching_overview can be used to compare the results from several toolmatchin_check sheets.

Hint:

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